



$$F(\eta) = e^{-\eta} \left(\frac{\alpha}{1+\alpha} \right)$$

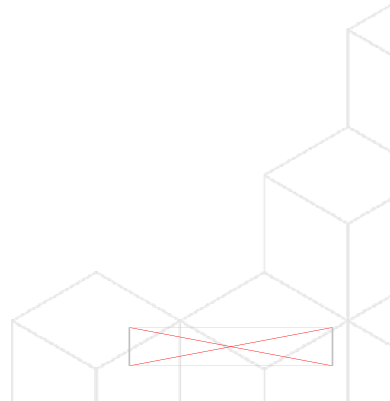
SerDes RX Discussion

Face-to-Face Meeting with Lightmatter — August 11, 2026

CDNS

Topics

- BER Target
- Channel Modeling Approach
- Link Budget
- Inverter-Based Front-End Interface with Cadence EIC
- Interleaved RX Topology for Power and Performance Optimization
 - Comparator Sampling Aperture Width
 - Leveraging 224 Gb/s PAM4 LR Building Blocks
 - Mueller-Müller CDR



BER Target

Product Concepts and Key PPA Metrics

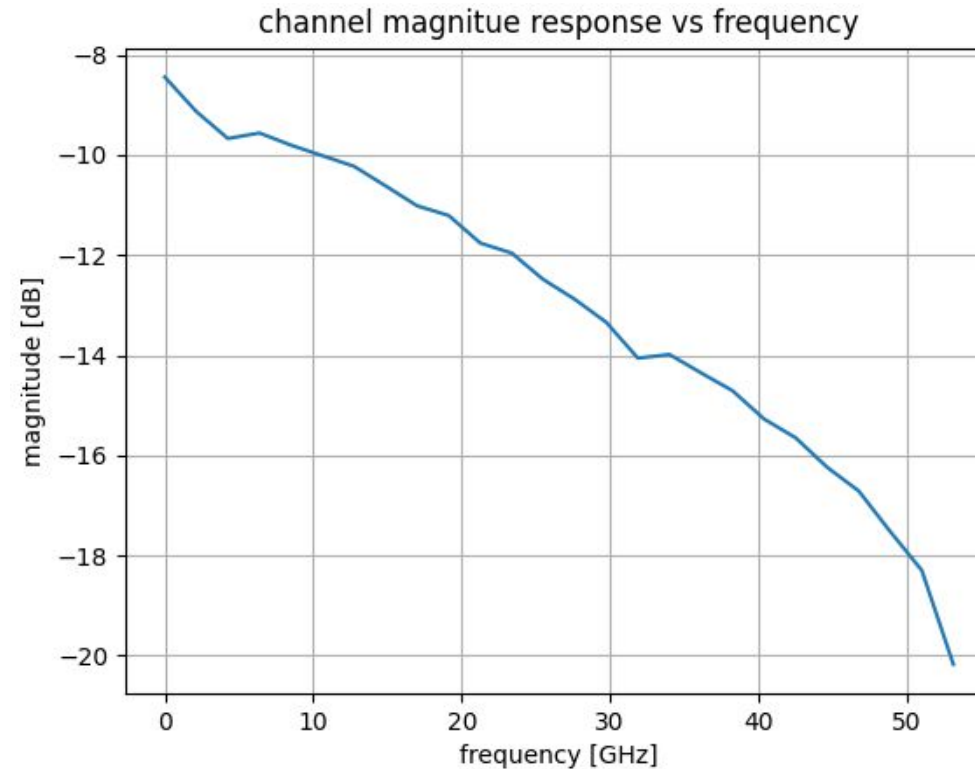
LIGHTMATTER PROPRIETARY AND CONFIDENTIAL

	Electrical Standards Compliant DR		OCI MSA DWDM	
	DR 200G	DR 400G	Gen 1: DWDM 200G	Gen 2: DWDM 400G
Line Rate [Gbps]	212.5	425	53.125 - 64	106.25 - 128
Modulation	PAM4	PAM4	NRZ	NRZ
Port BW [Gbps]	200	400	200	400
Unidi/Bidi	Unidi	Unidi	Bidi	Bidi
BW/fiber [Gbps]	200	400	400	800
SerDes Power [pJ/bit] target	2	2	1-2	1-2
PIC Power [pJ/bit] target	2	2	1	1
Laser Power [pJ/bit] target	1	1	1	1
Area/lane [mm^2] target	0.5	?	0.15-0.2	0.15-0.2
Max BER	1.00E-06	1.00E-06	1.00E-12	1.00E-12
Max symbol errors/RS FEC frame	10	10	3	3
Latency	>100ns	>100ns	20-50ns	20-50ns

Source: *LM+CDNS_Next Gen SerDes.pdf*

Assumed raw BER target: 1e-12

Channel Modeling



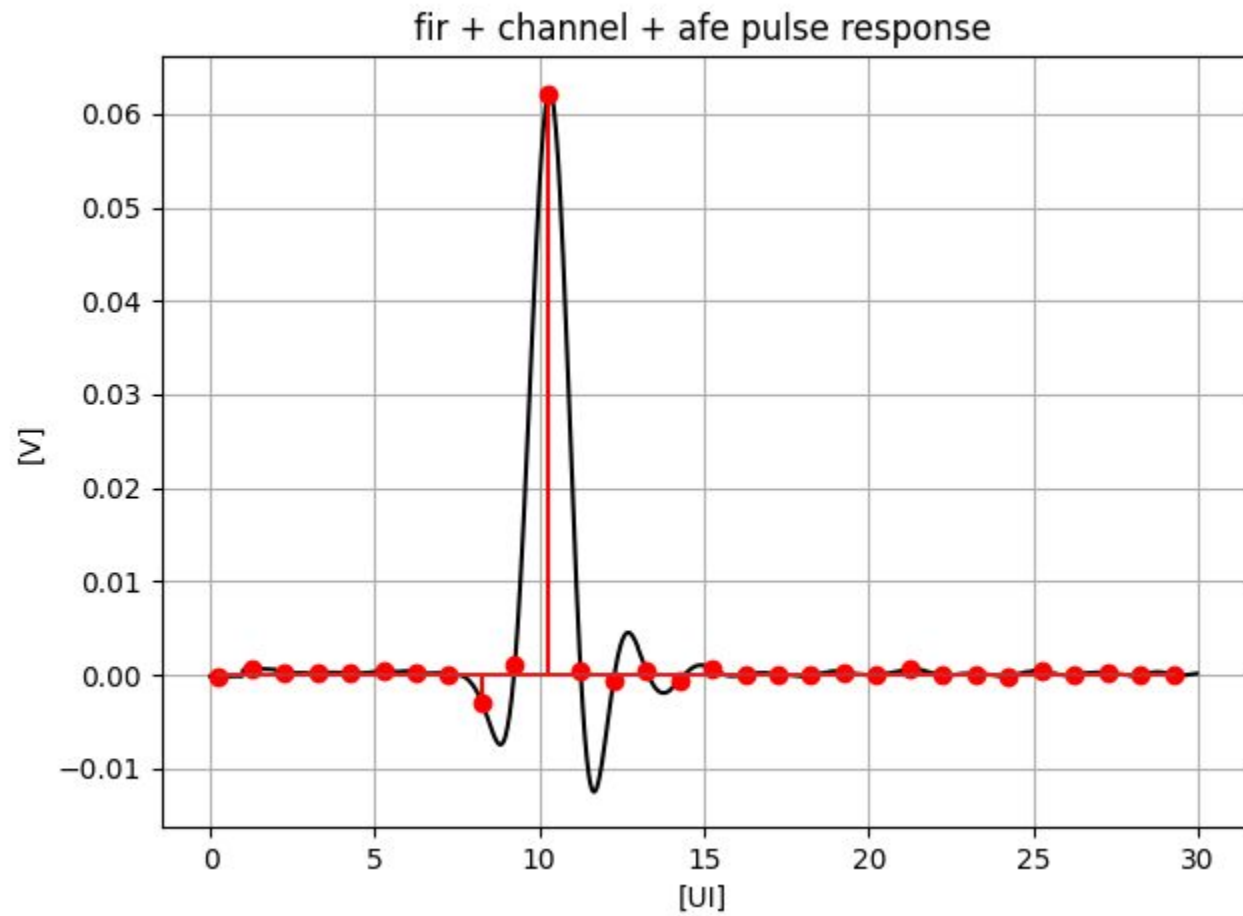
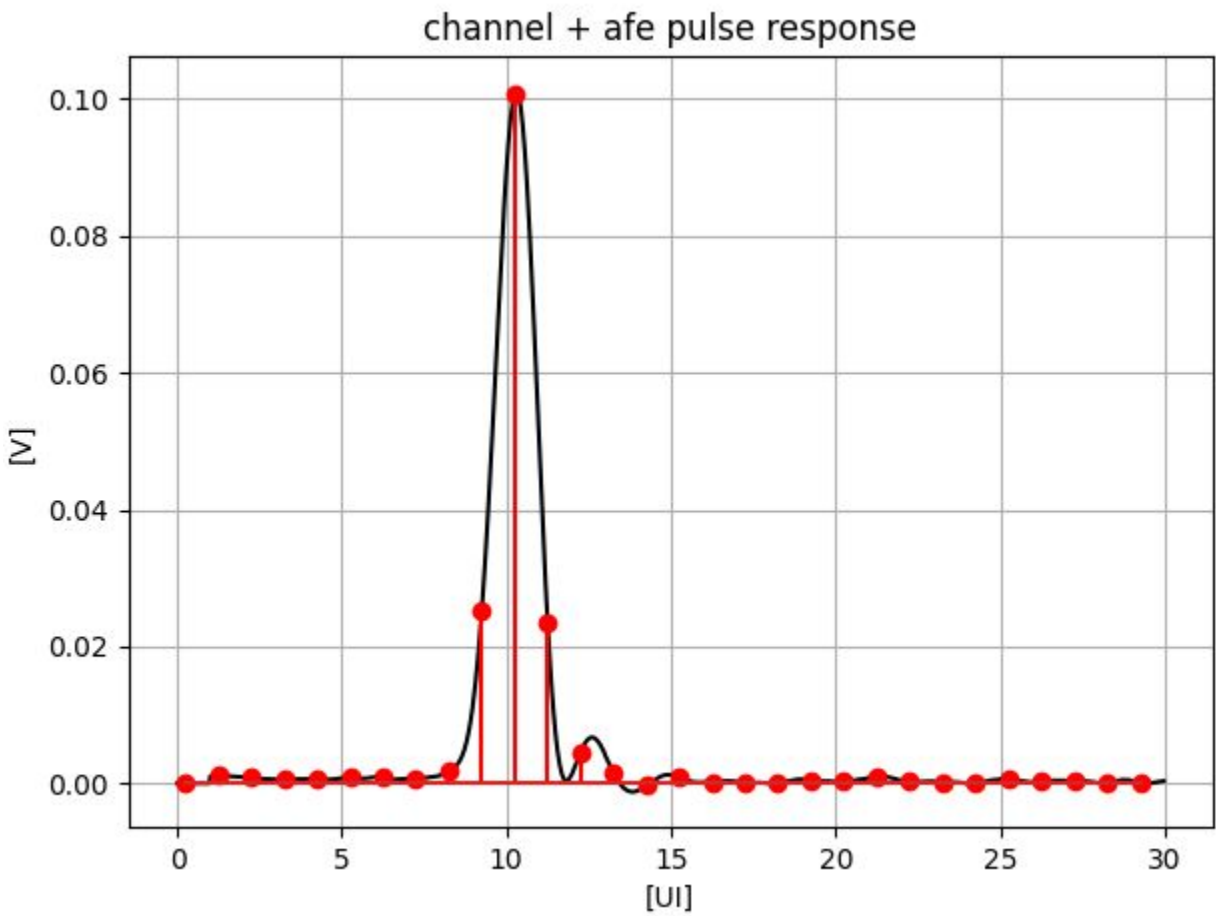
- The current budget calculation uses an LM channel response derived from PRBS measurements, including the full PIC signal path.
- For improved accuracy, future analysis should use schematic-level Spectre models, or at minimum, full-channel S-parameter data.



Link Budget

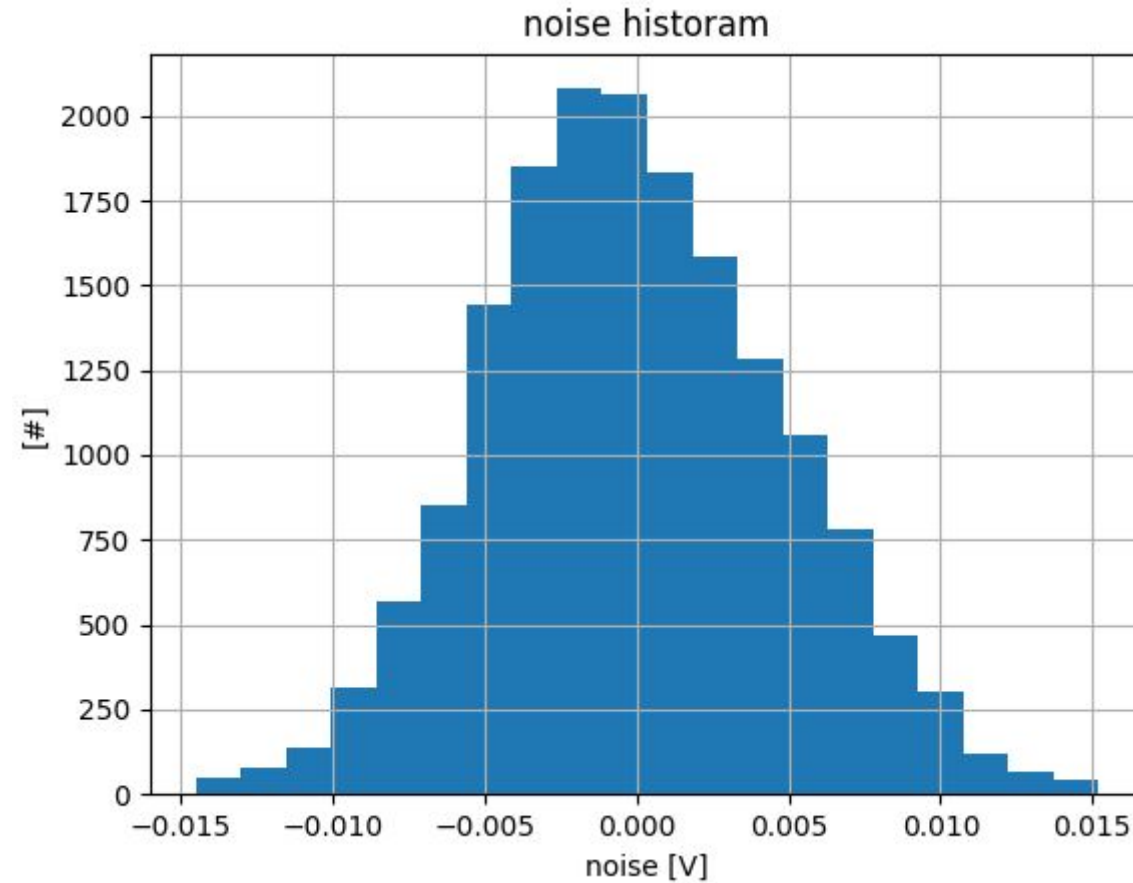


Pulse Response of the Equalized Channel with a 3-Tap TX FIR Equalizer



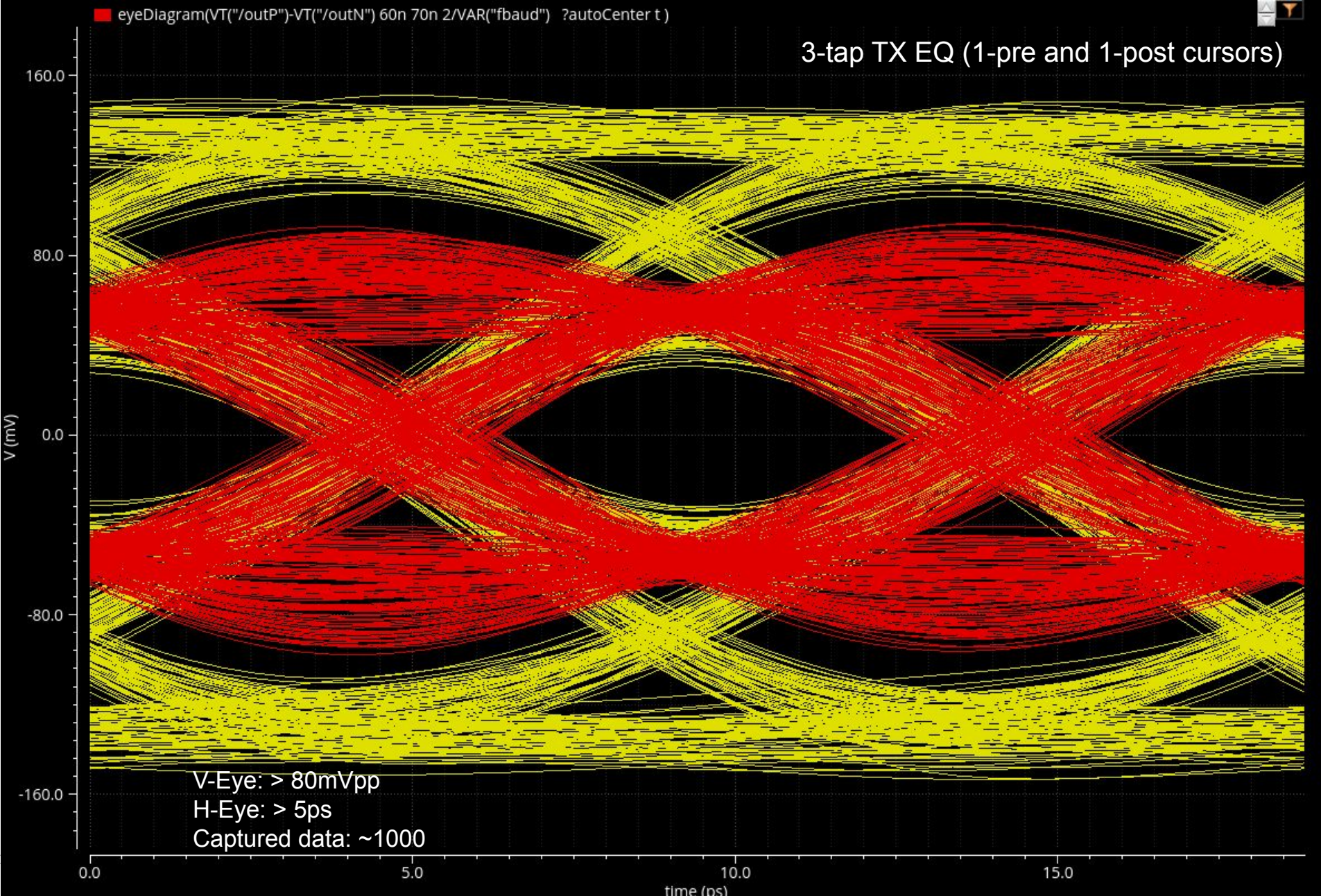
TX FIR Coefficients (Pre/Main/Post): -0.15 / 0.69 / -0.16

RMS Noise Estimation from PRBS Data



- Estimated RMS Noise: 4.84 mV
- Second- and third-order nonlinear components have been removed prior to noise estimation

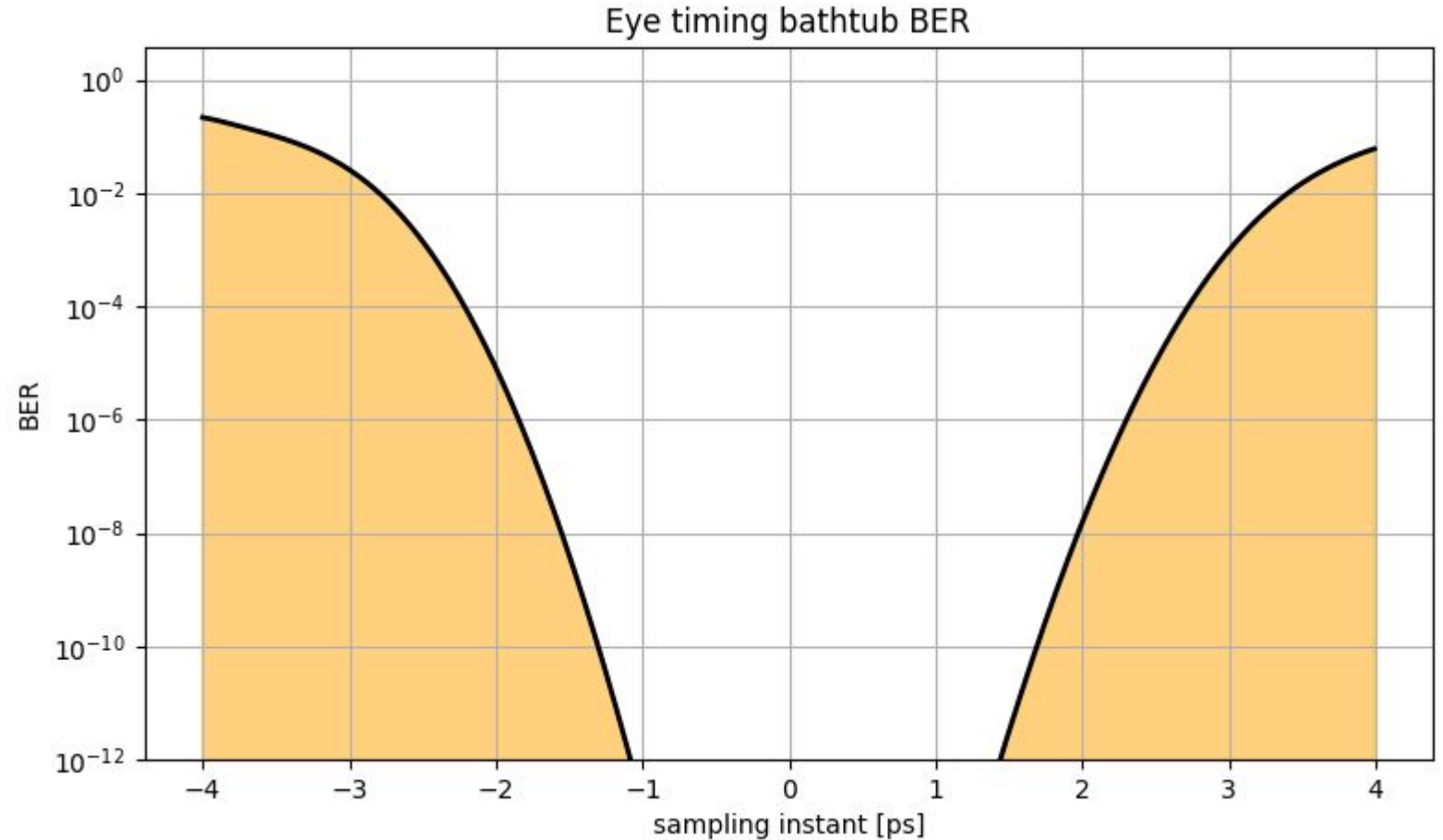
Eye Diagram Comparison: With and Without 3-Tap TX FIR



BER Estimation (RX Differential Swing ≈ 120 mVpp)

System Simulation Conditions

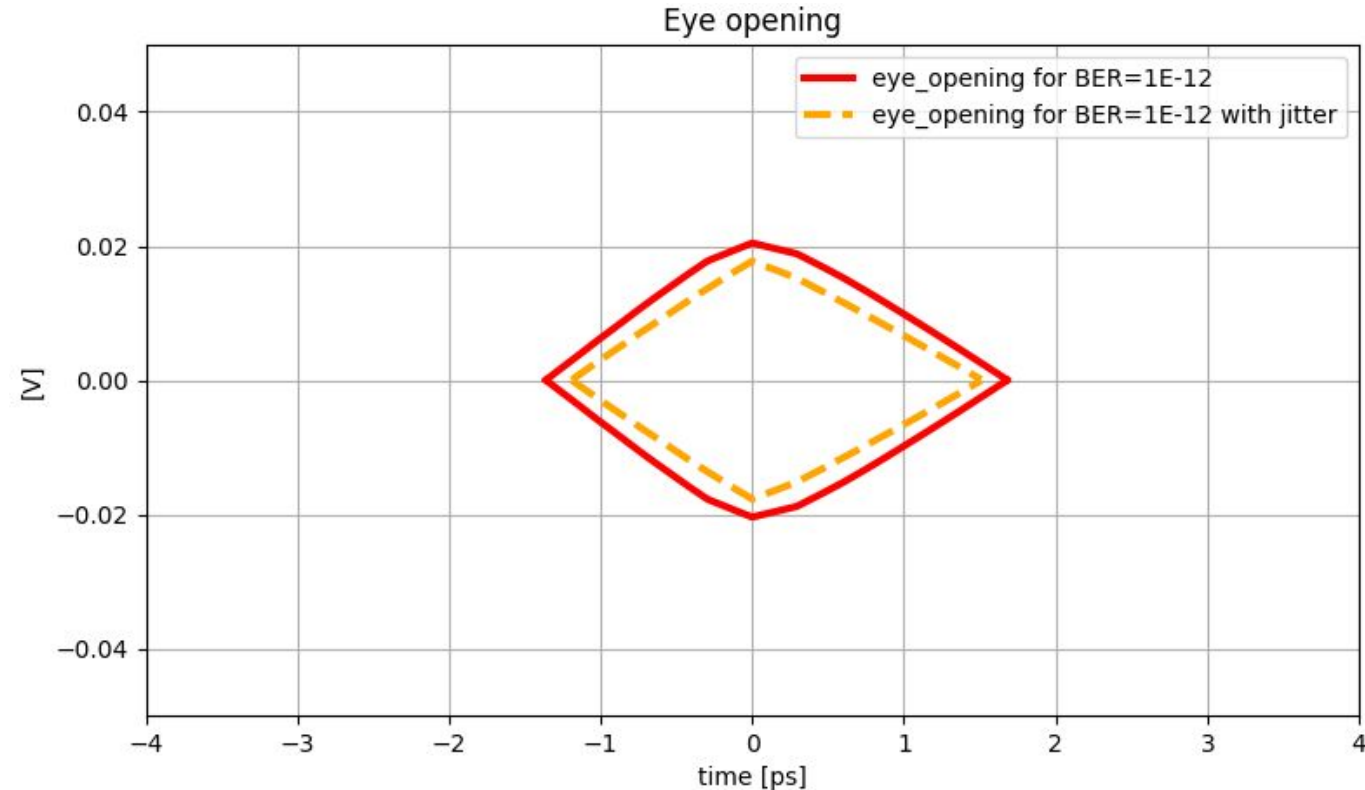
- RX differential swing: 120 mVpp
- RJ: 100 fs RMS (TX and RX)
- DJ: 100 fs RMS (TX and RX)
- RMS noise: 5 mV
- 3-tap TX FIR enabled



Estimated Eye Opening with a 3-Tap TX FIR Equalizer

System Simulation Conditions

- RX differential swing: 120 mVpp
- RJ: 100 fs RMS (TX and RX)
- DJ: 100 fs RMS (TX and RX)
- RMS noise: 5 mV
- 3-tap TX FIR enabled



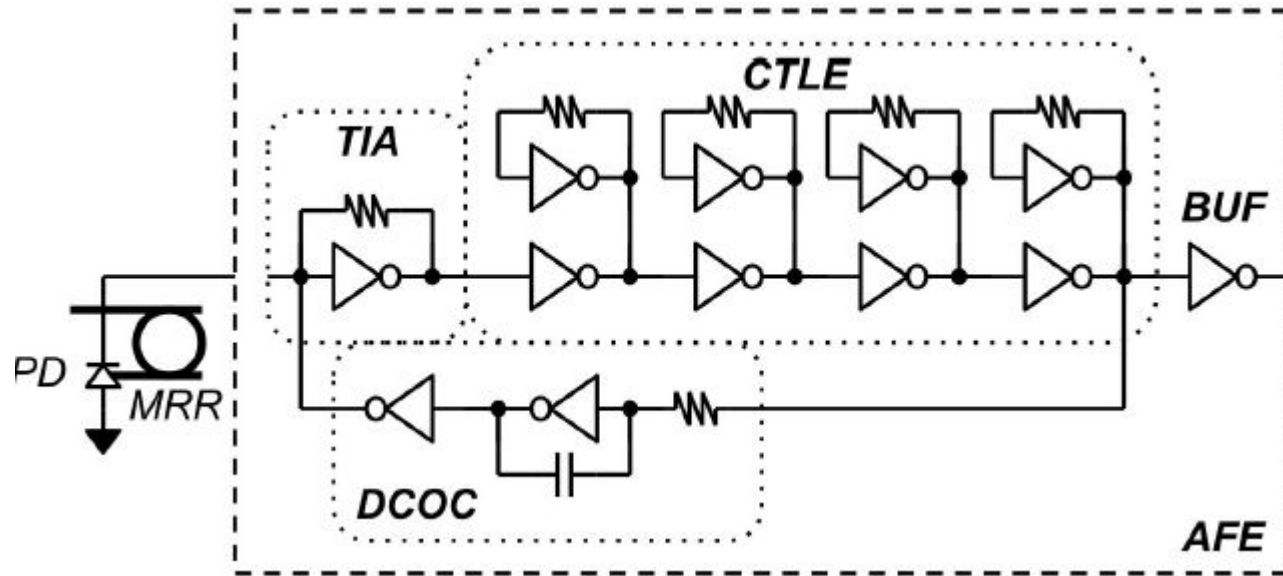
- 106.25 Gb/s eye opening of ~ 40 mV and 0.28 UI at $\text{BER} = 1 \times 10^{-12}$ with 3-tap TX FIR
- RX EQ with CTLE and/or FFE desirable to add more margin and adapt for PVT variability



Inverter-Based Front-End Interface with Cadence EIC



Inverter-Based Front-End Interface with Cadence EIC



Reference: R. Baghdadi et al., "Monolithically Integrated Microring Transmitter and Receiver for High-Density 3D Co-Packaged Optics", OFC 2025

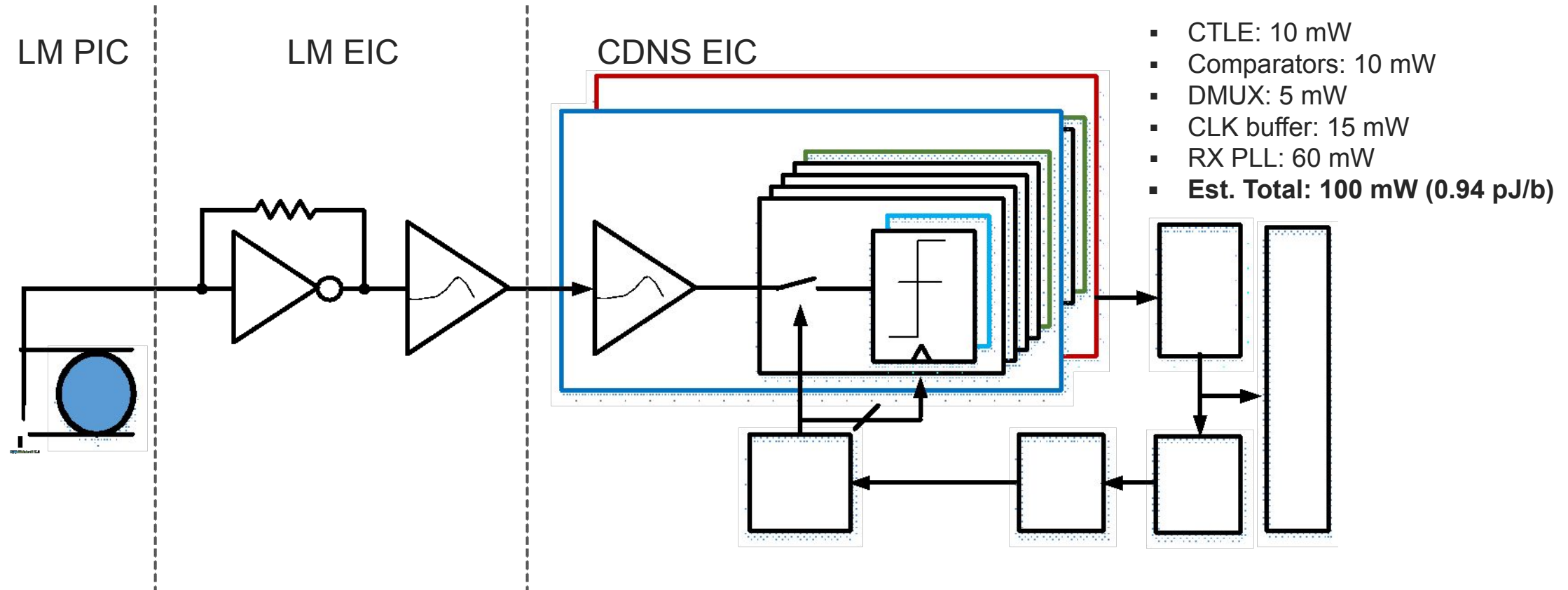
- It would be beneficial to use the actual LM TIA/CTLE implementation as the basis for the input buffer and CTLE design within the Cadence EIC.



Interleaved RX Topology for PPA Optimization

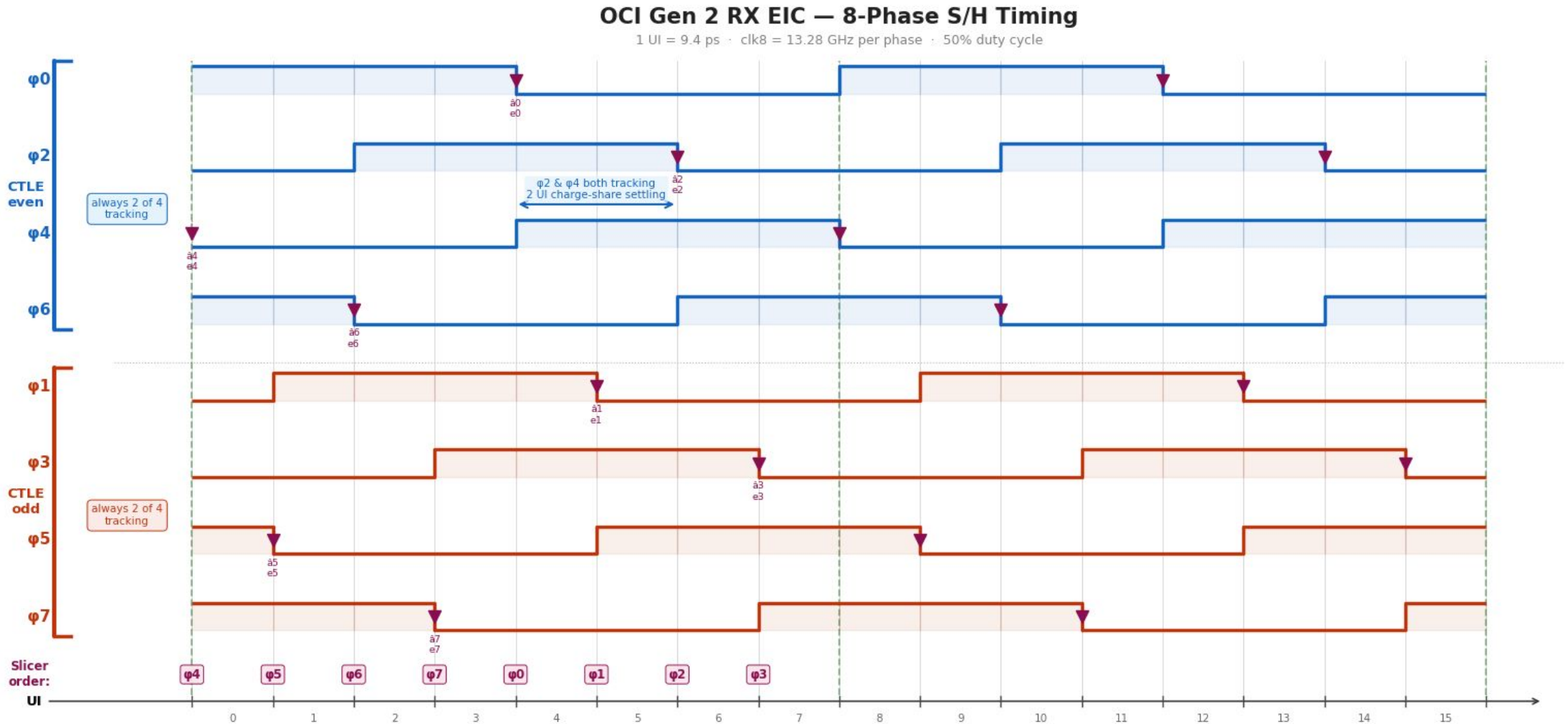


Proposed Interleaved RX Architecture



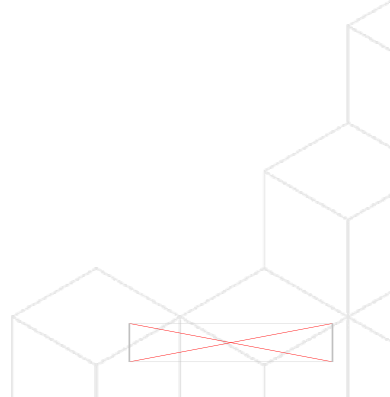
- Interleaved RX with S/H reduces CTLE loading, mitigates sampling-aperture-induced ISI, and allows the comparator to operate at a noise-optimized speed.
- Even/odd interleaving extends bandwidth and reduces charge-sharing-induced ISI.
- Baud-rate Mueller-Muller CDR requires only an error comparator, no edge comparator.

8-Phase S/H Timing

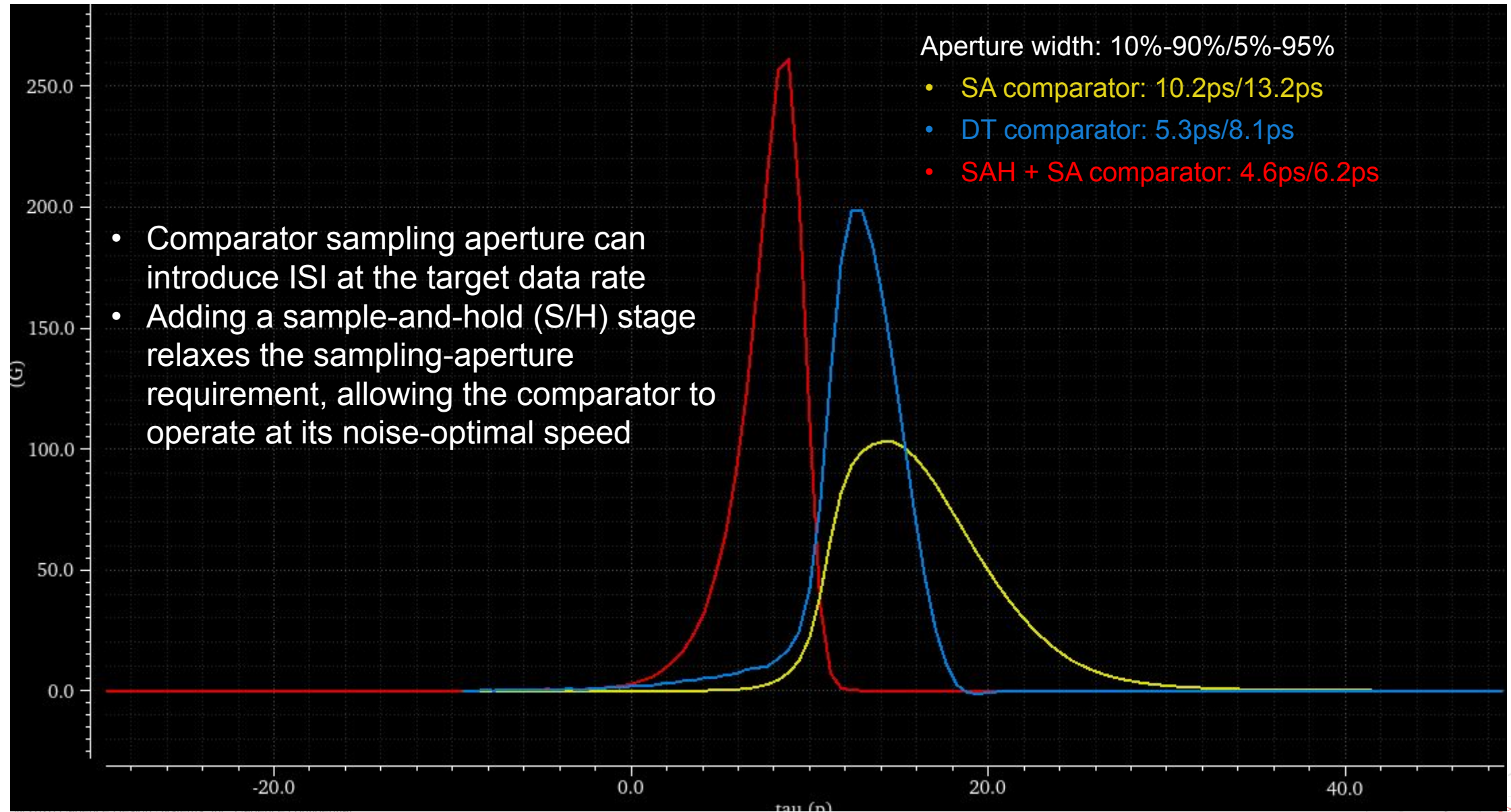


- At any given time, only two of the four even slices and two of the four odd slices are actively tracking, therefore limiting CTLE loading.

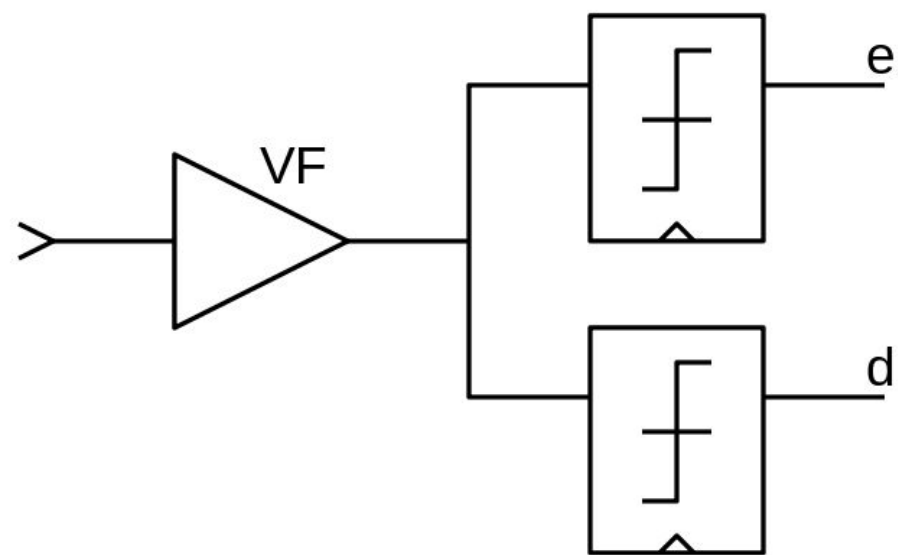
Comparator Sampling Aperture Width: Why an Interleaved RX?



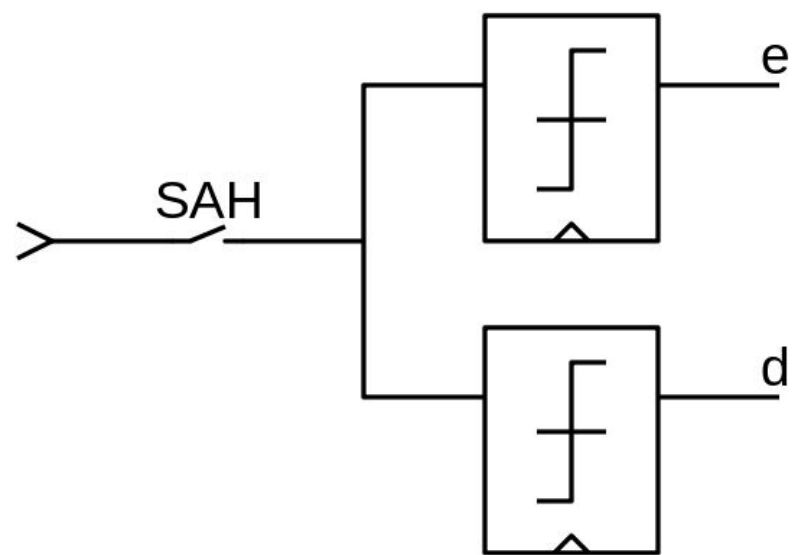
Comparator Impulse Sensitivity Function (ISF) Comparison



RX Sampler Architecture Comparison



(a) Nyquist Voltage Follower + Comparator



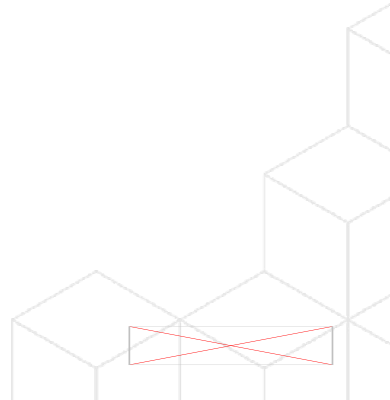
(b) S/H + Comparator

Metric	Architecture (a)	Architecture (b)
Clock Buffer Power	Lower	Higher
Datapath Power	Higher	Lower
Signal Bandwidth	Lower	Higher
Noise Performance	Higher	Lower

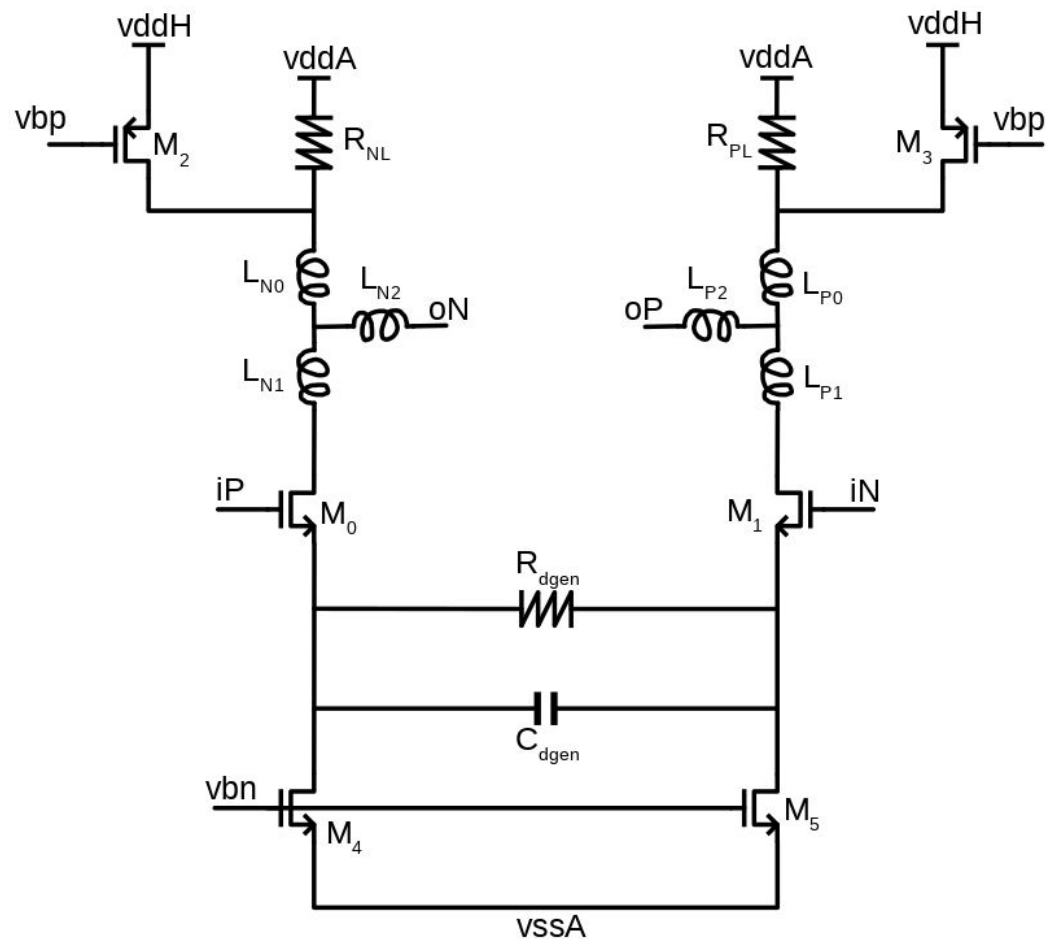
Architecture (b), an interleaved RX with S/H, reduces sampling-aperture-induced ISI while improving bandwidth and noise performance.



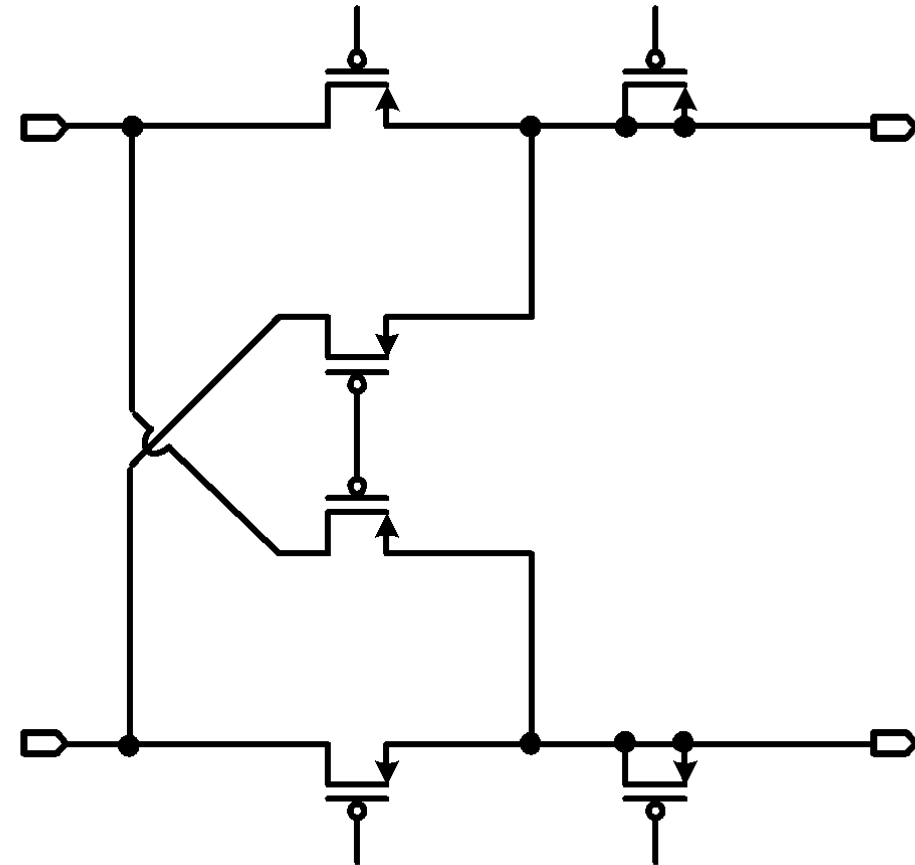
Leveraging Silicon-Proven 224 Gb/s PAM4 LR Building Blocks



CTLE and S/H Switch Optimized for 224 Gb/s PAM4 LR IP

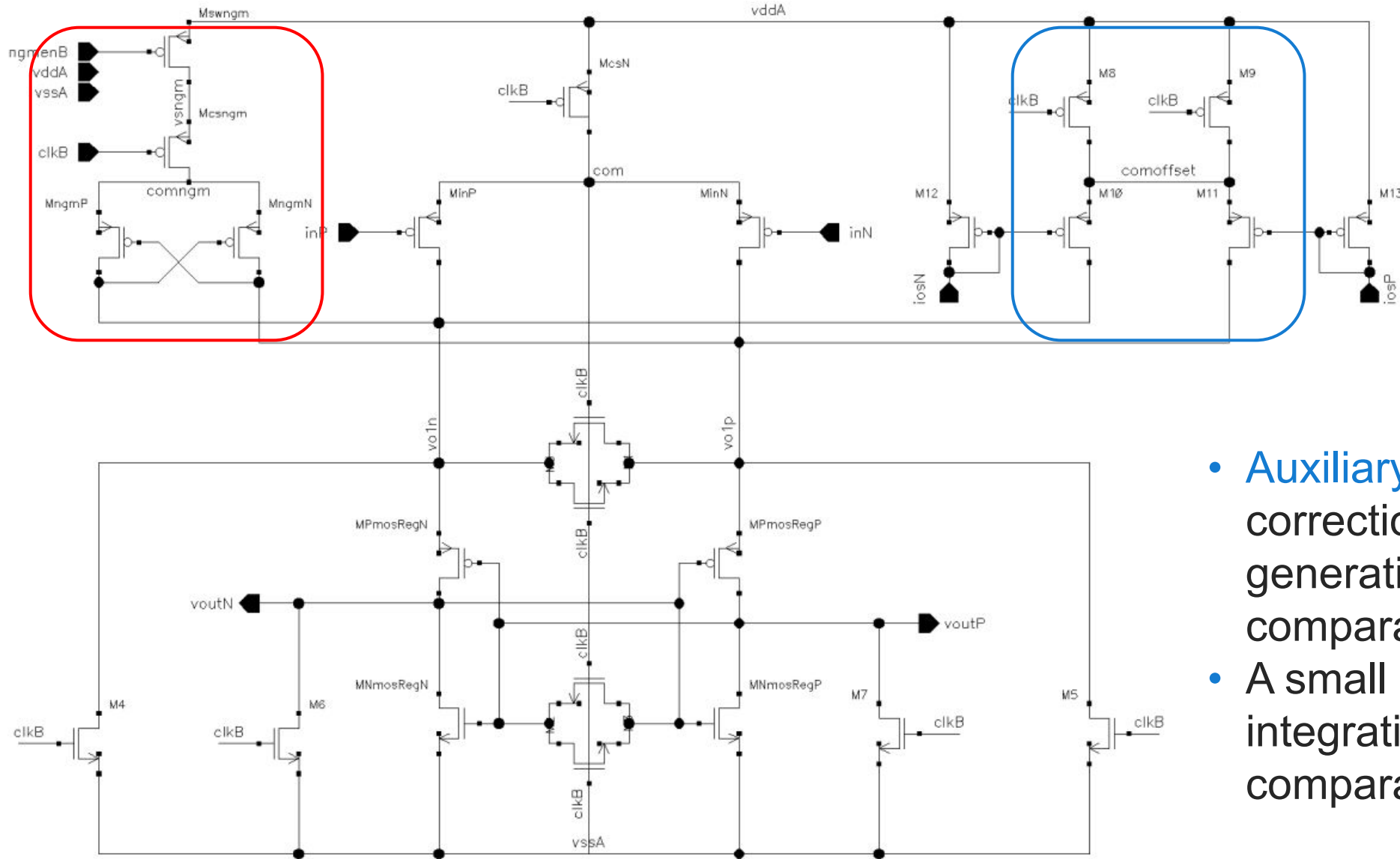


- CML CTLE with RC degeneration and a 3-inductor T-coil, providing flexible frequency-response shaping



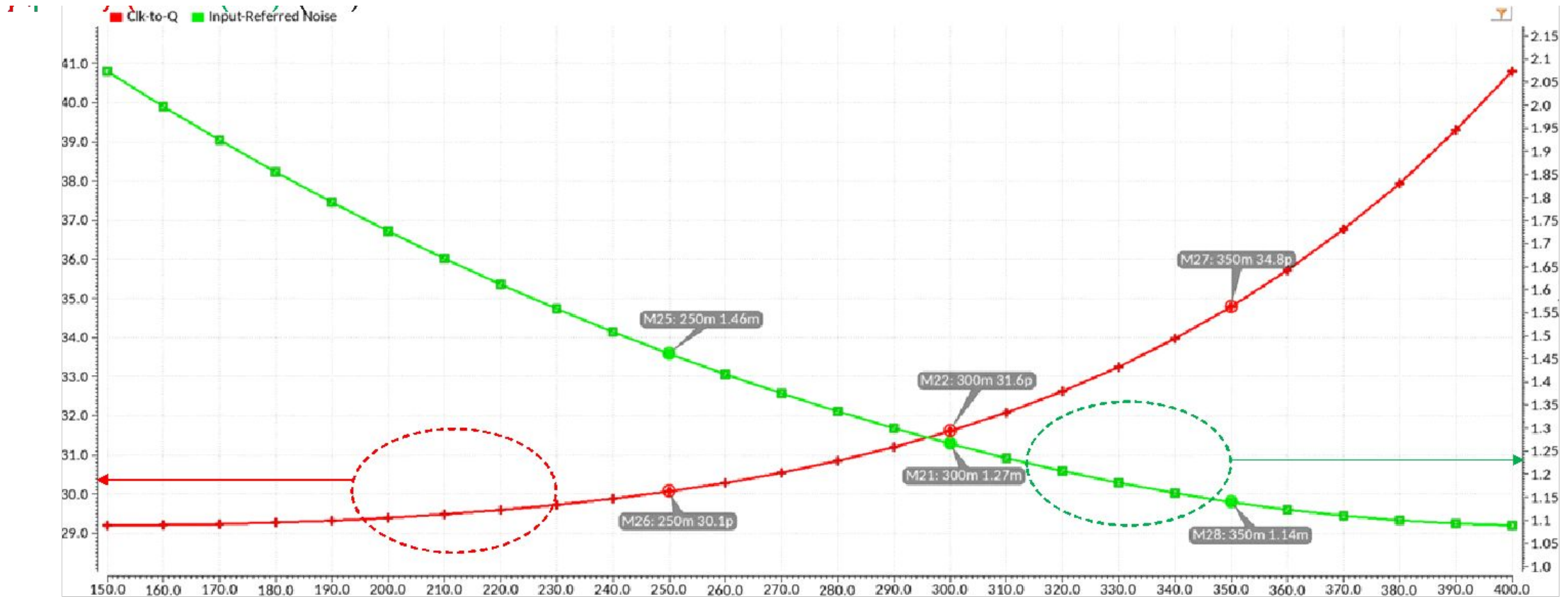
- pFET sampling switch designed to minimize clock feedthrough and mitigate off-state signal disturbance

StrongARM Comparator Optimized for 224 Gb/s PAM4 LR IP



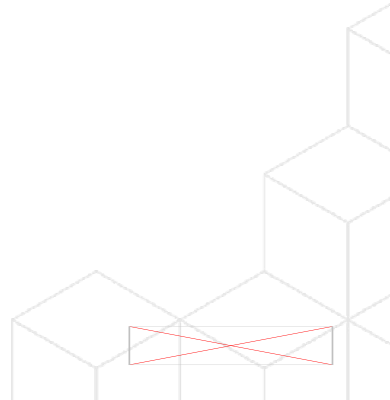
- Auxiliary input pair for offset correction and reference generation for the error comparator
- A small **negative-gm cell** at the integration node to speed up comparator operation

Clk-to-Q Delay and Input-Referred Noise vs. Input Common-Mode Voltage



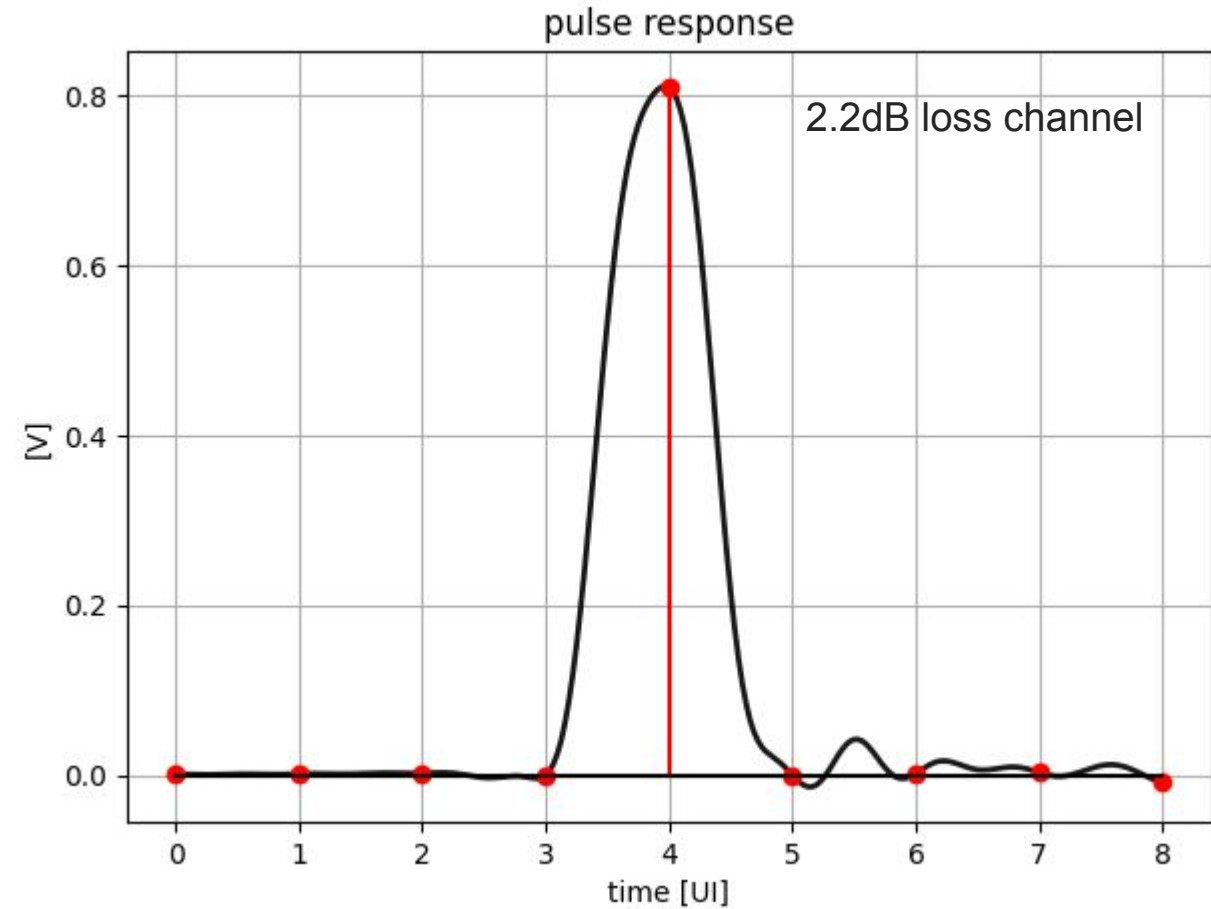
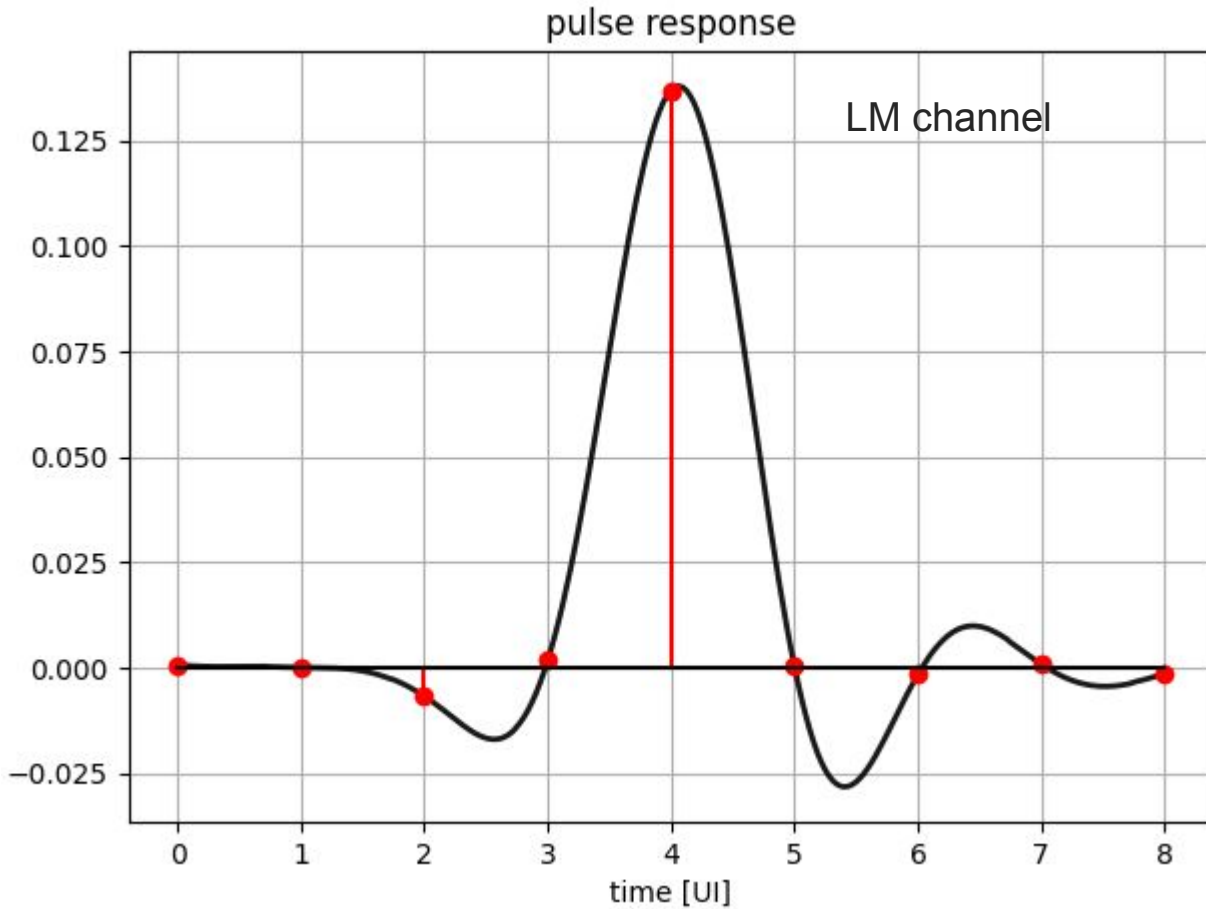
- The S/H switch relaxes the comparator speed requirement, enabling operation across a wider input common-mode voltage range

Mueller-Müller CDR

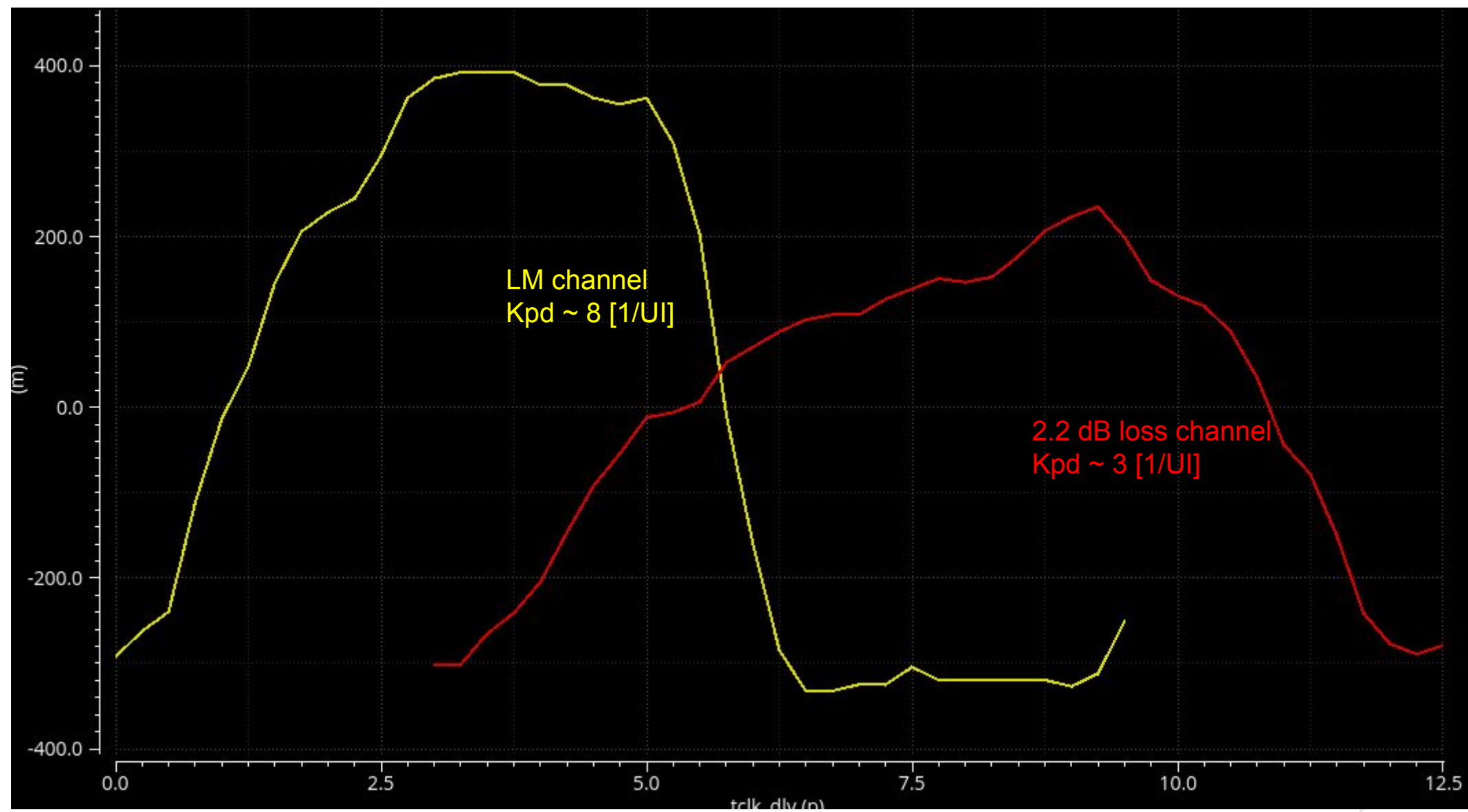


Pulse Response Comparison: LM Channel vs. 2.2 dB Loss Channel

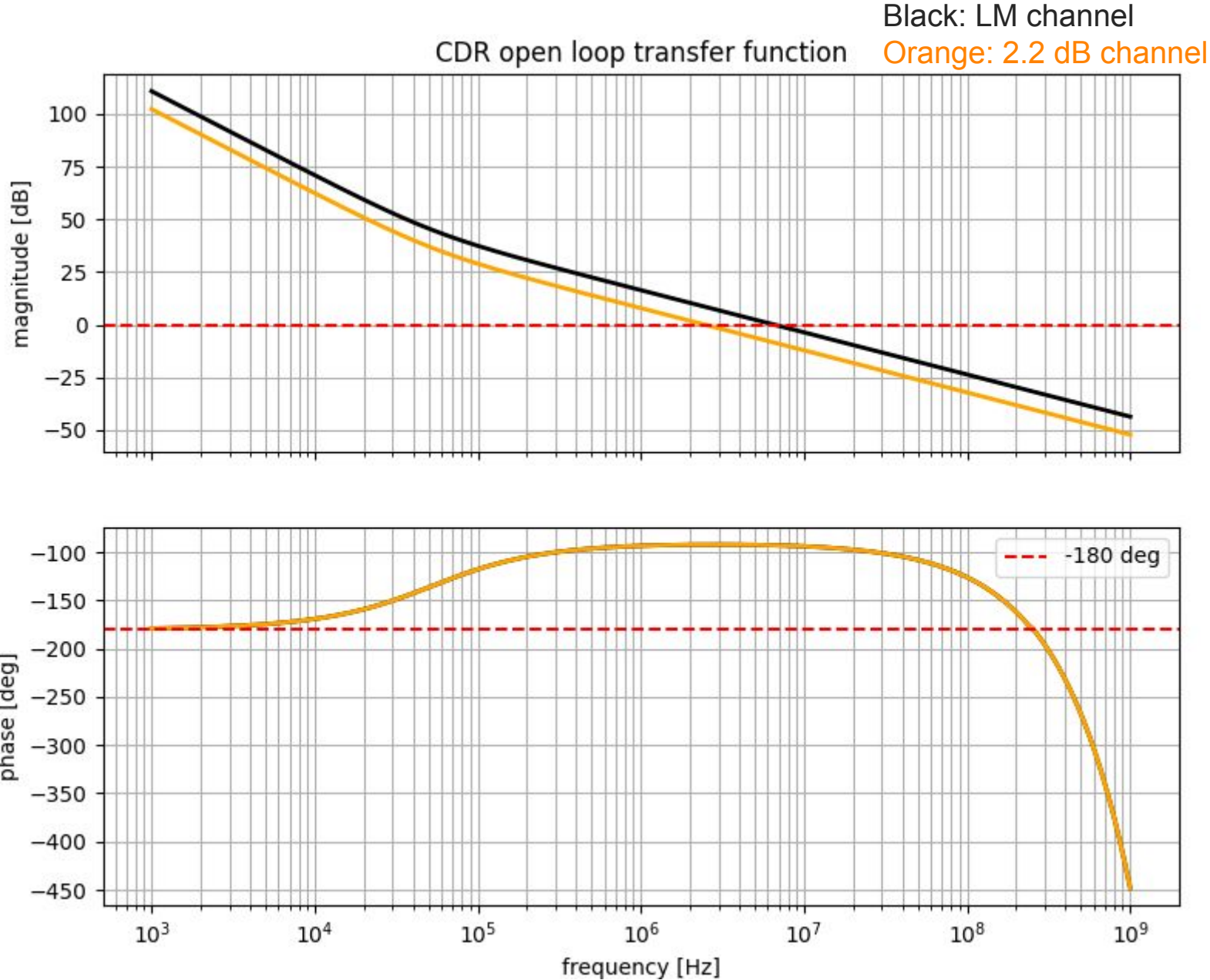
- Pulse responses shown after applying a 3-tap TX FIR equalizer



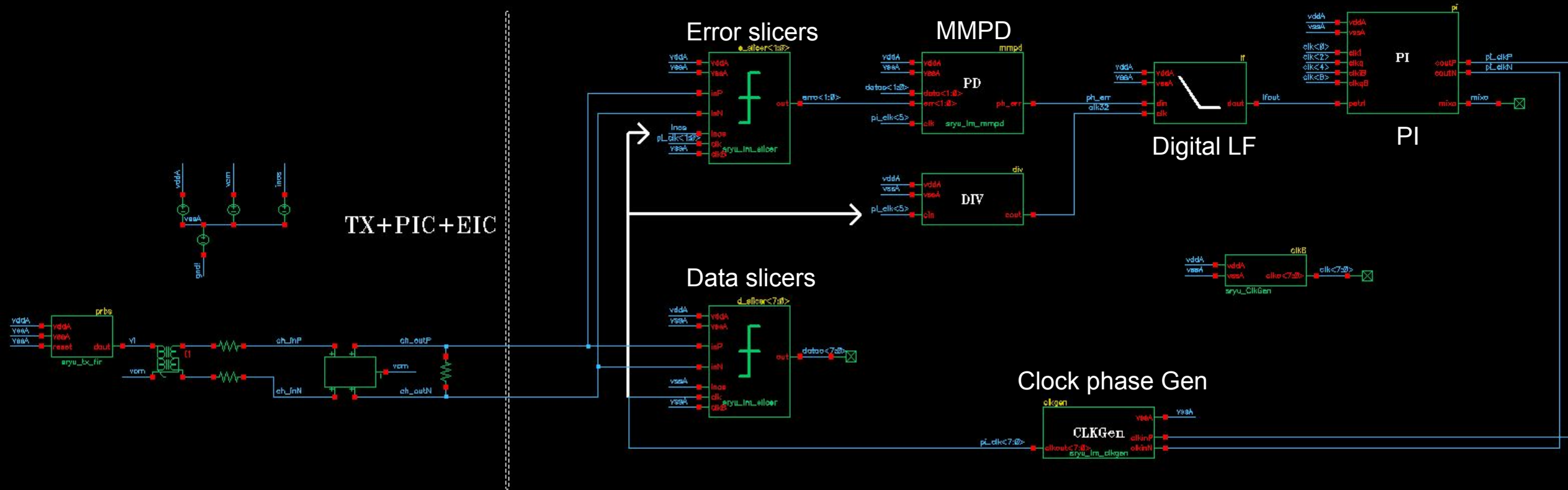
Sign-Sign MMPD Timing Functions



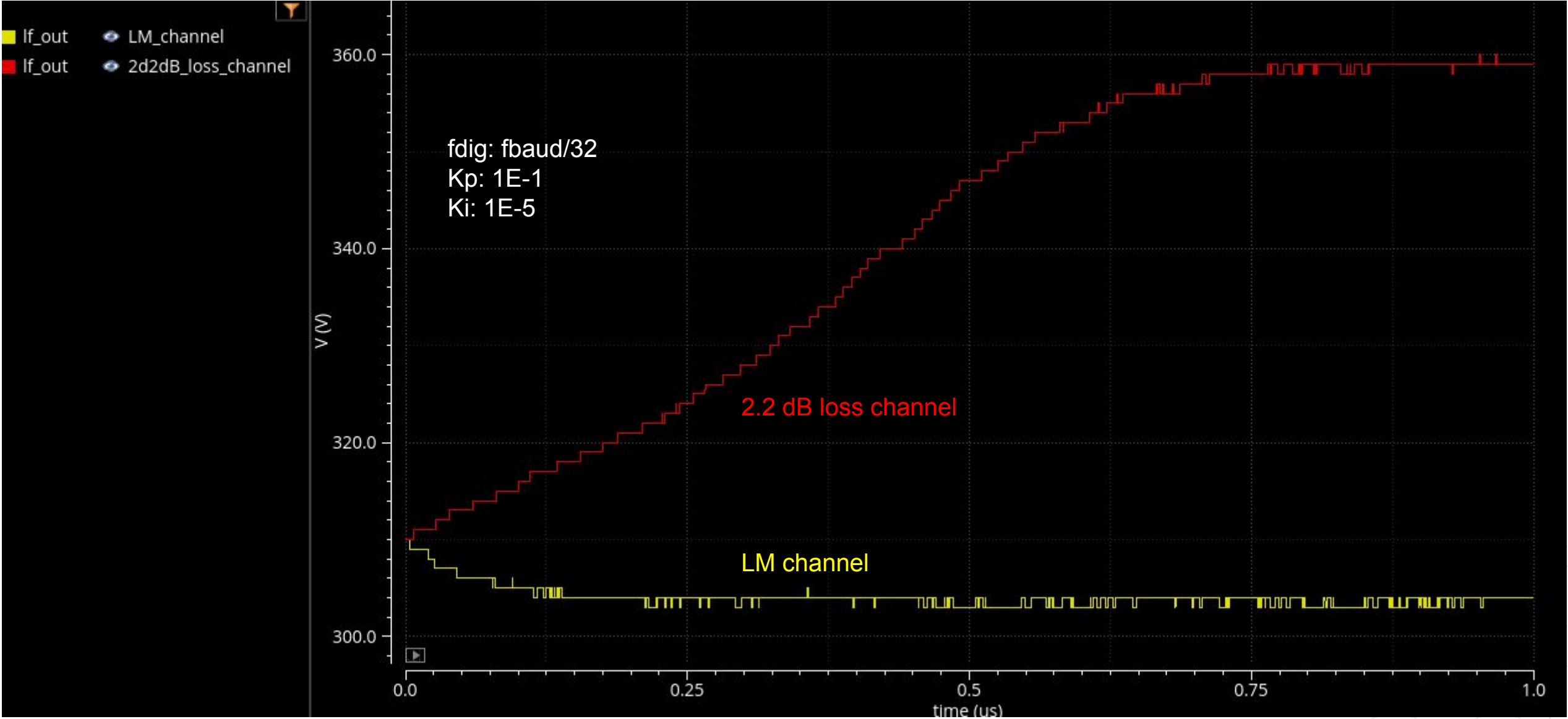
CDR Loop Transfer Functions



MMCDR Testbench (Verilog-A)

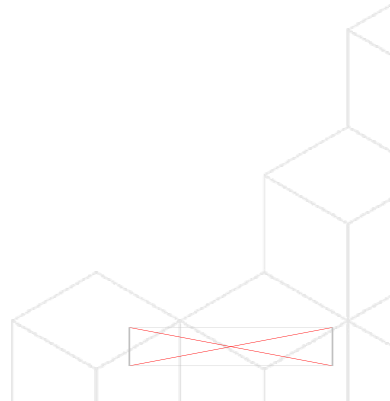


CDR Time Domain Simulation (Verilog-A)



Questions

- Can we confirm the target raw BER requirement?
- What is the preferred approach for modeling the PIC data path? Can S-parameter models, schematics, or Verilog/behavioral models be provided to support margin analysis?
- Using the existing LM Gen2 TIA/CTLE implementation as the basis for the Cadence EIC input buffer and CTLE design would be highly beneficial. Can additional details, schematics, or design documentation for the current Gen2 implementation be shared?





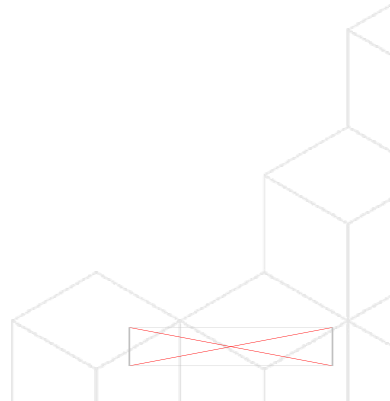
SerDes TX

F2F meeting @ LM 08/11/2026

CDNS

Outline

- Cadence 128Gb/s NRZ TX
- Initial Simulation Results
- Q&A



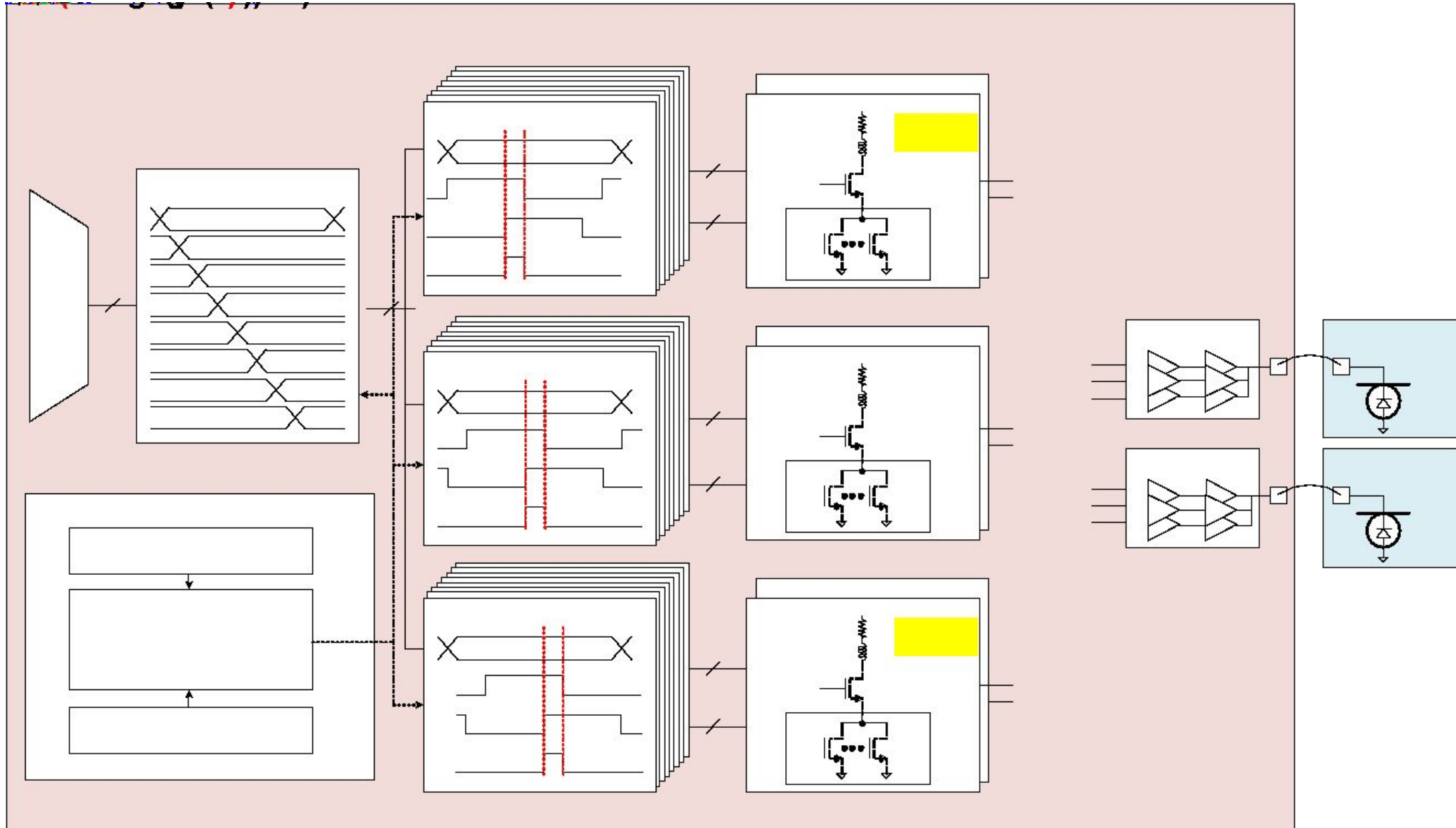


Cadence 128Gb/s NRZ TX

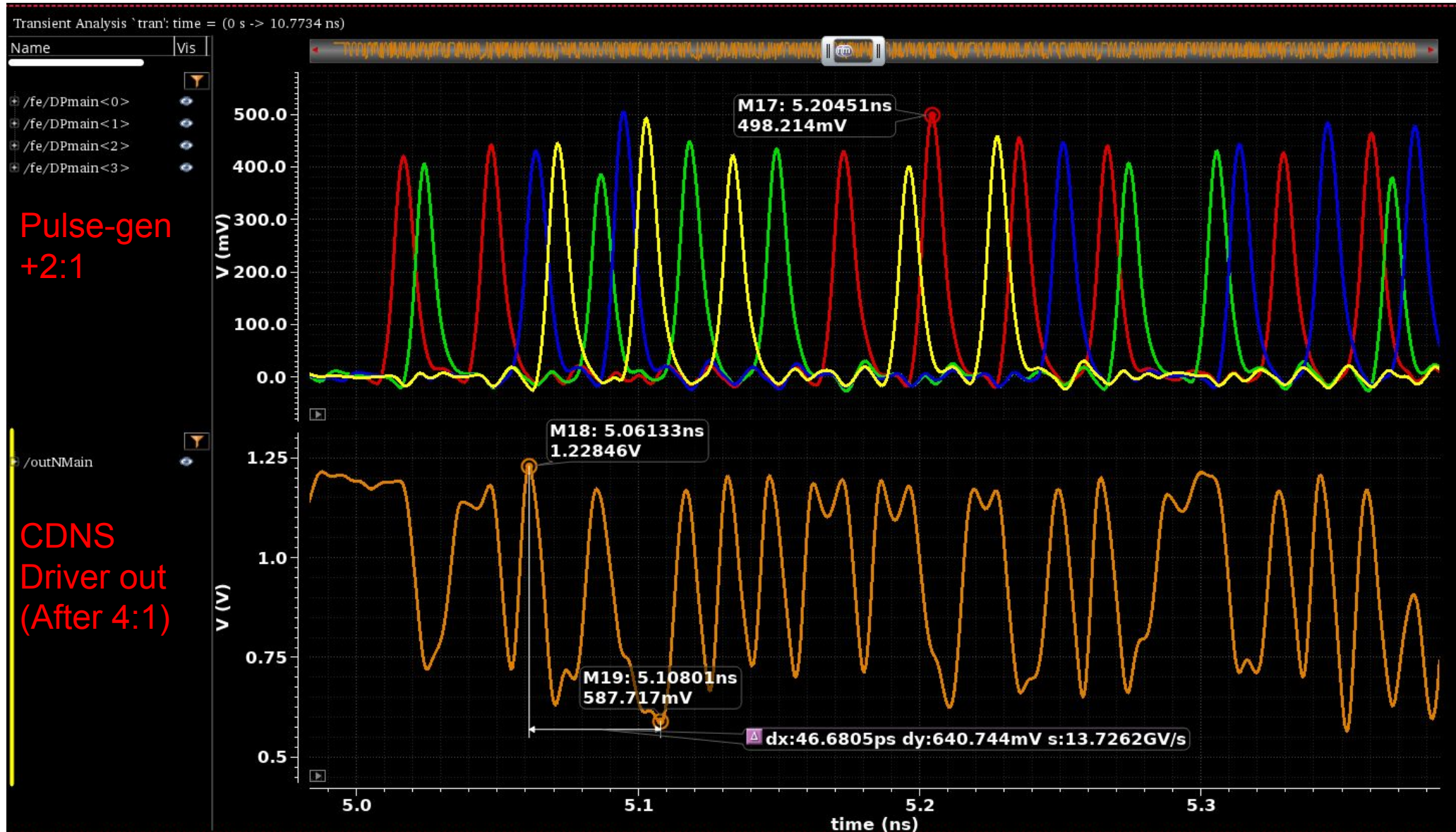
{Modified from Silicon-Proven 224Gb/s PAM-4 IP}



TX Architecture (3-paths: Pre/Main/Post) – can be merged early/late



Waveforms for TX Core Blocks- Pulse-Gen & Driver (@128Gb/s)



*WIP
-Different pulses
-Will be fixed



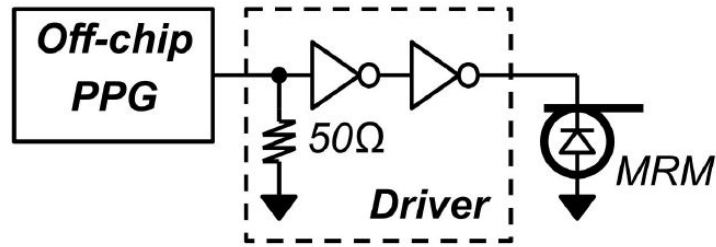
Initial Simulation Results

{Assumption: Inverter-type/CML-type Final Driver}



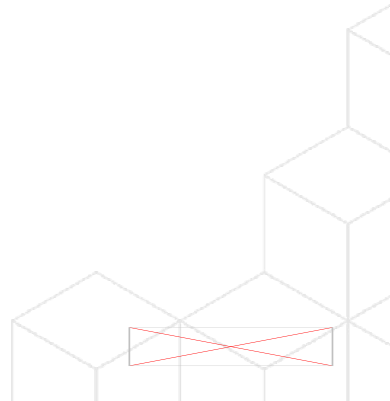
Reference from Lightmatter

Transmitter Architecture



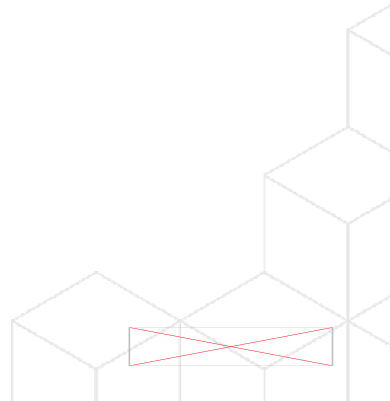
- Monolithically integrated driver and MRAM in 45nm CMOS
- Driver Details:
 - **Inductorless** Single-ended CMOS inverter with 50Ω input termination
 - Two-stage buffer chain
 - Output swing: 1.3V pp (56Gbps)
- TX Die Area: $2.8 \times 10^{-3} \text{ mm}^2$

[Ref-OFC,2025] Reza Baghdadi et al.,

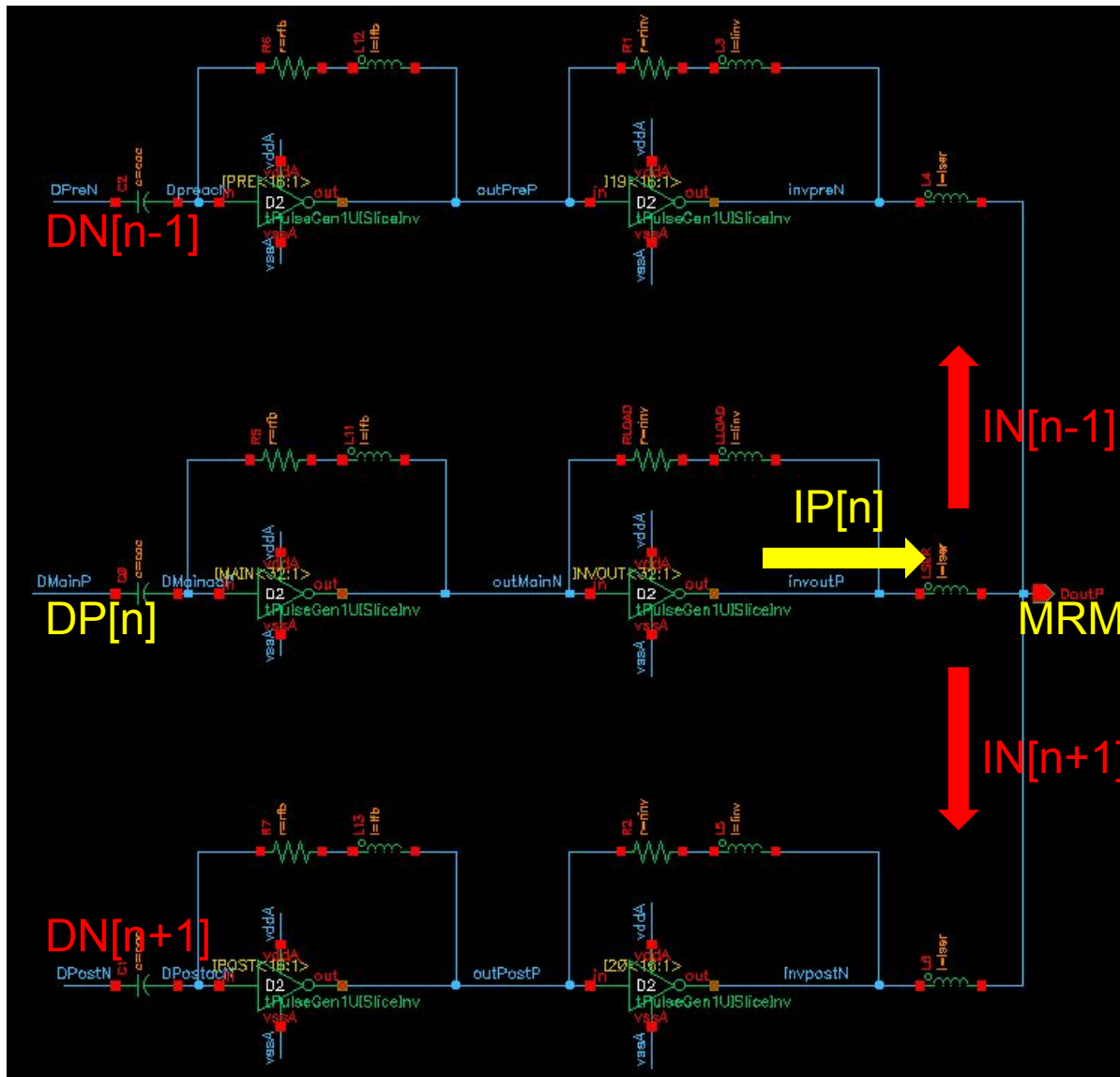


Initial Assumptions

- Tested with PRBS9 pattern
- IO supply used for LM Driver
- **Unlikely** but assumed for initial simulations...
 - Load model for MRM: 10fF (with R=200ohm, BW~80GHz)
- Simulation: CDNS Serializer + 2 types of driver candidates
 - CMOS-type LM driver
 - CML-type LM driver
 - **Note: Preliminary sim**



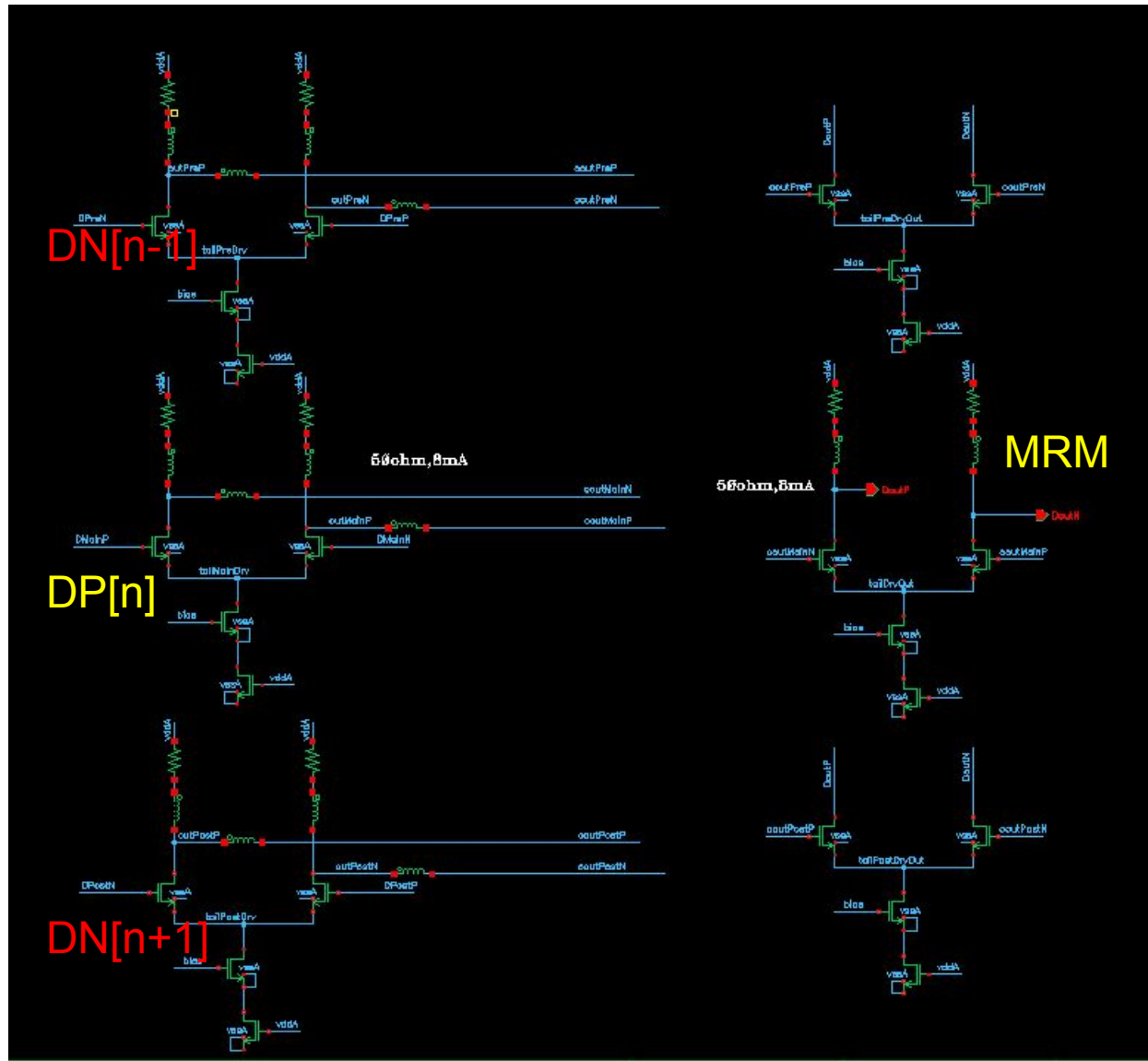
1.LM:CMOS-type Driver



*Notes for CMOS-type driver

- Can get higher swing than CML generally
- It may need level conversion (ac-coupling capacitor)

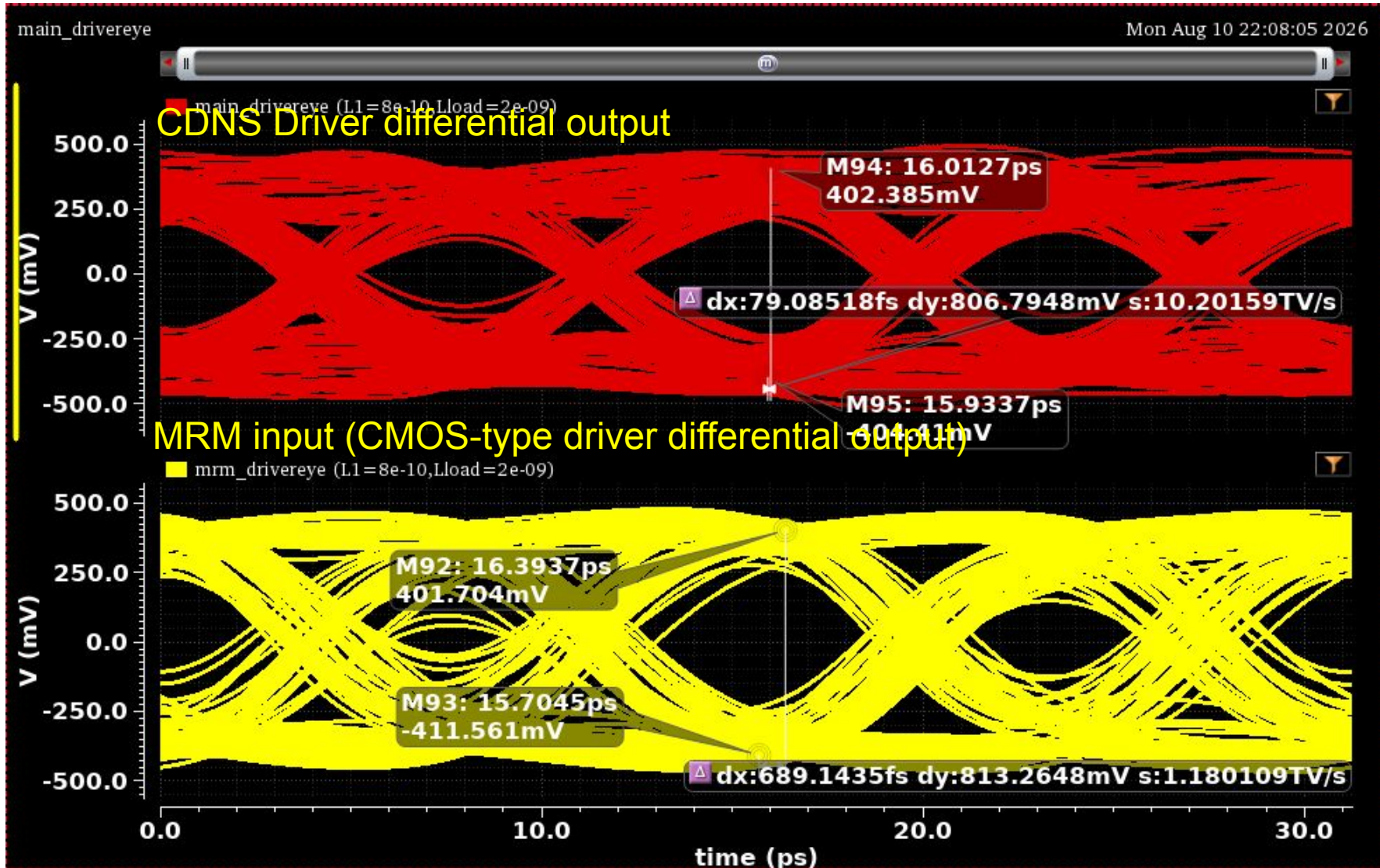
2.LM:CML-type Driver



*Notes for CML-type driver

- BW determined by RC
- Swing: lower swing than CMOS but generally less PVT sensitive

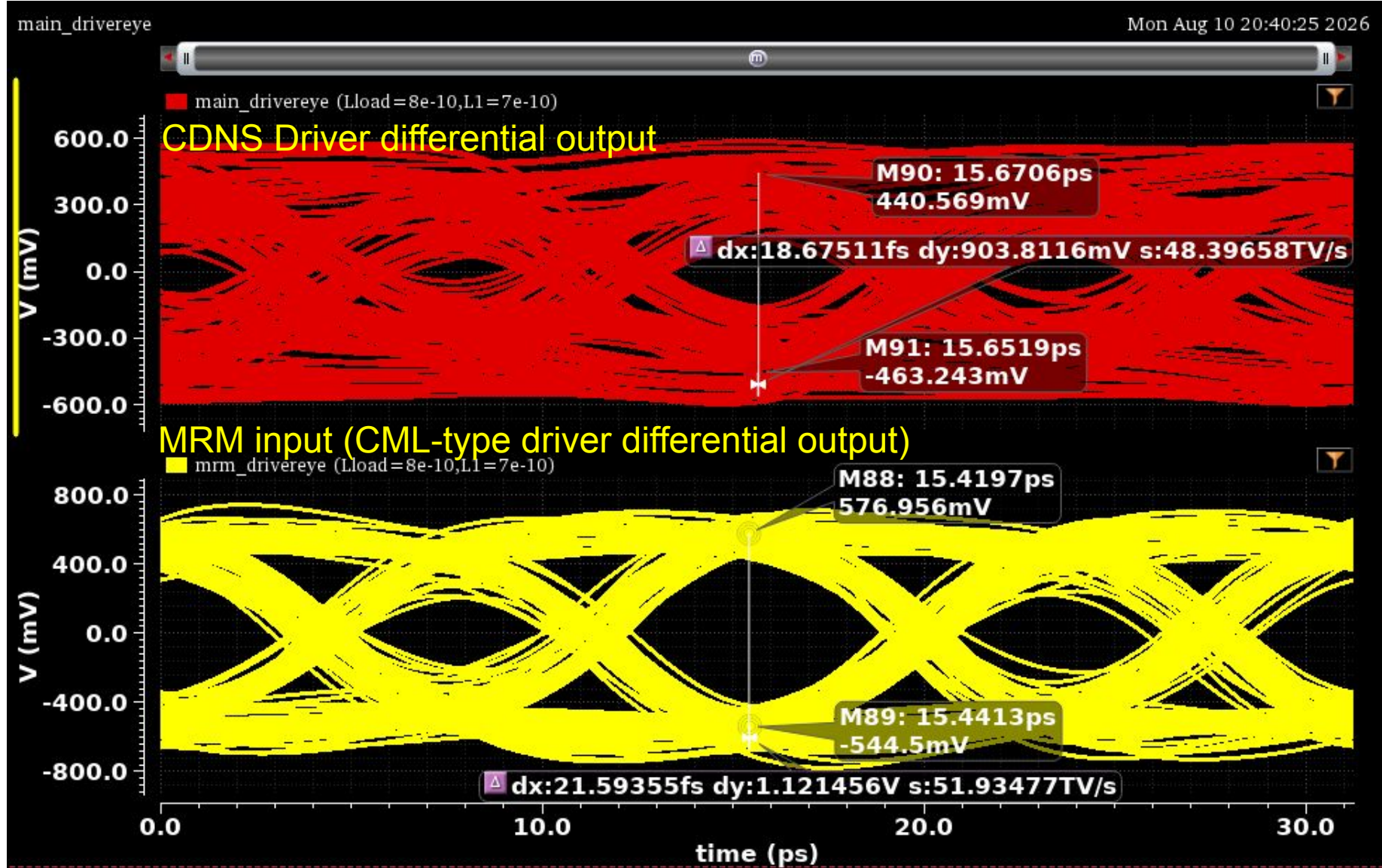
128Gb/s NRZ Eye Diagrams – CMOS-type (Pre/Post: 5% ON)



*WIP:
-Have to run ac
simulation for
optimization

*WIP:
-Even/odd asymmetric,
eye closure
-Will be fixed

128Gb/s NRZ Eye Diagrams – CML-type (Pre/Post: OFF)



*WIP:
-Have to run ac
simulation for
optimization

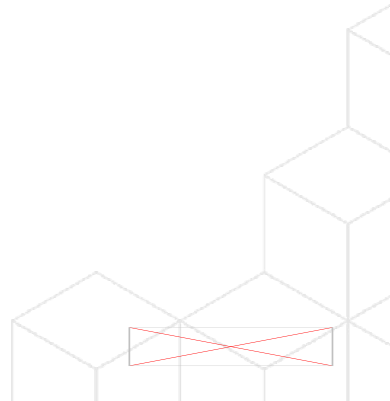
*WIP:
-Even/odd
asymmetric
-Will be fixed

Power Number

Power(mW) at 128Gb/s NRZ		
3-tap FFE(50%/100%/50%) Differential		CDNS (Linnet)
Initial Sim	Driver	8.6
	Pulse Gen	11.8
	Retimer	2.6
	txTopCkBuf	4.5
	tClkBufTop	45.7
*Estimated	Backend	8.4
	TXPLL	72.3
Total Power(mW)		153.9
Efficiency(pJ/b)		1.2

1. Can save 50% if single-ended OK
 2. Can save more if max FFE can be lower

→ May amortize TXPLL across lanes





Q&A



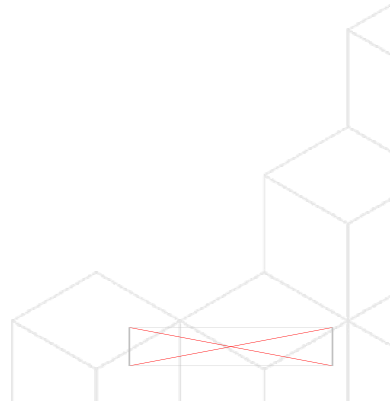
Questions

- Electrical-side

- Input spec. to LM TX? (i.e. voltage, capacitance, etc.)
 - Is LM driver located IO-voltage domain?
 - Do we need level conversion?
 - Other CDNS driver topology?
- Is it okay to have 3-paths (pre/main/post) to LM driver-side?
 - Target FFE(Pre/Post) maximum strength compared to Main?

- Optical-side

- What's the RC-model for MRM?
 - To run top-level simulation
- Is it single-ended or differential driven?
 - Okay to receive single-ended input from Cadence?



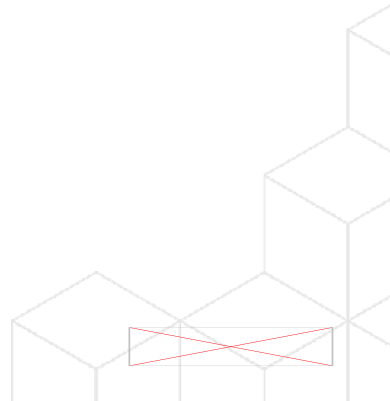
SerDes Top Level Items

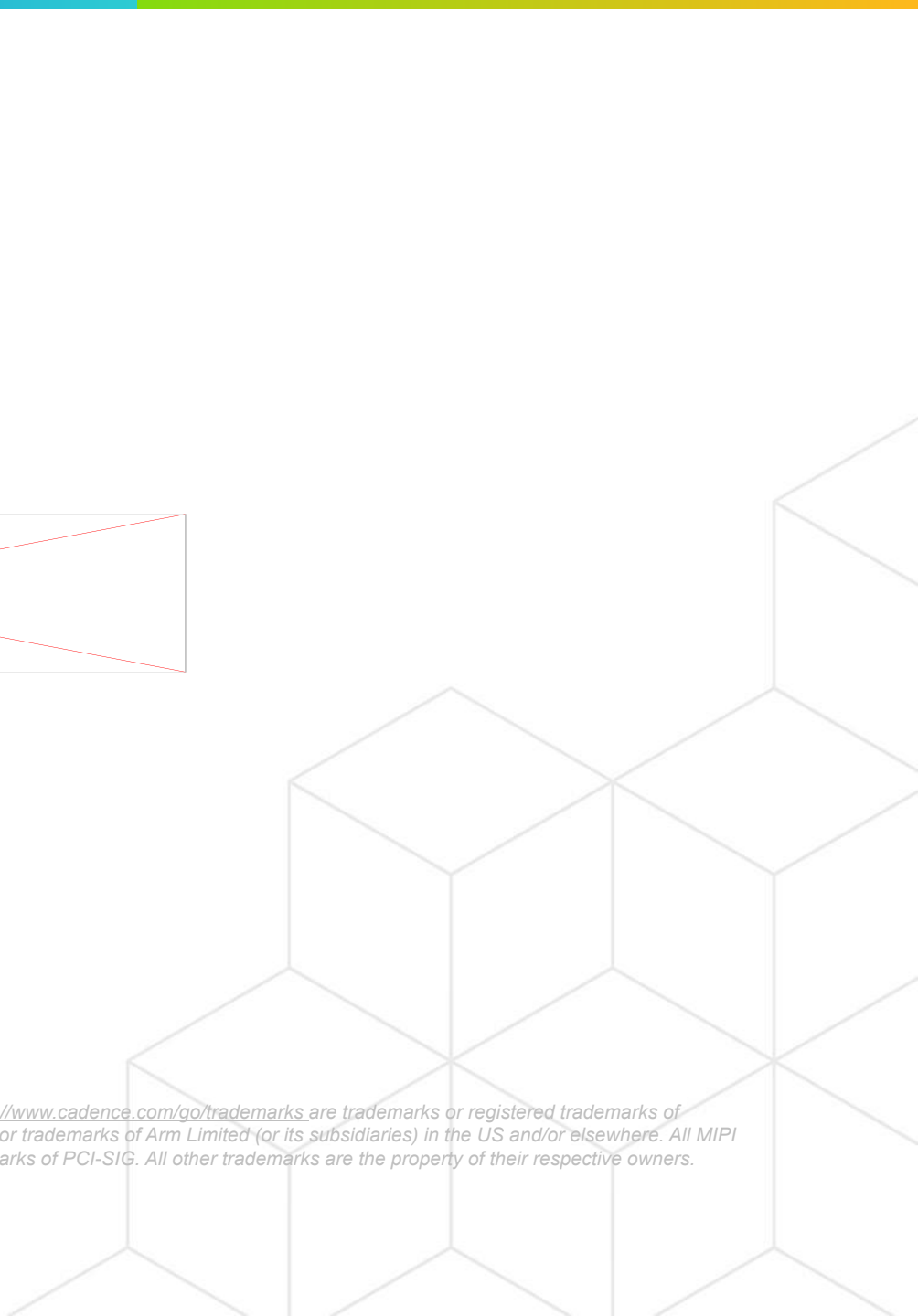
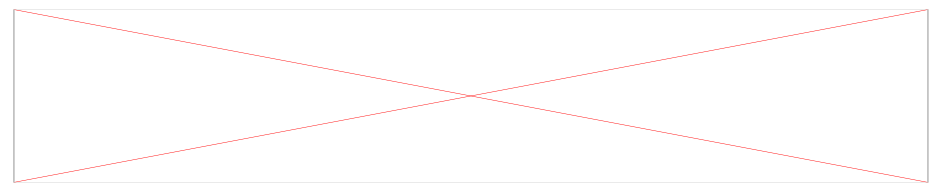
F2F meeting @ LM 08/11/2026

CDNS

Other Considerations

- Microcontroller required for some calibrations, e.g:
 - TX clock skew
 - Common mode calibration, if needed
- How to test electrical portion of design separately from optics?
- Plan to mostly reuse clocking chain from Kiwi (200G design)
 - Resonant clock distribution from common PLL to lanes
 - Aggressive power target may necessitate some work to reduce lane clocking power
- Bias blocks (bandgap, current mirrors) reused from Kiwi



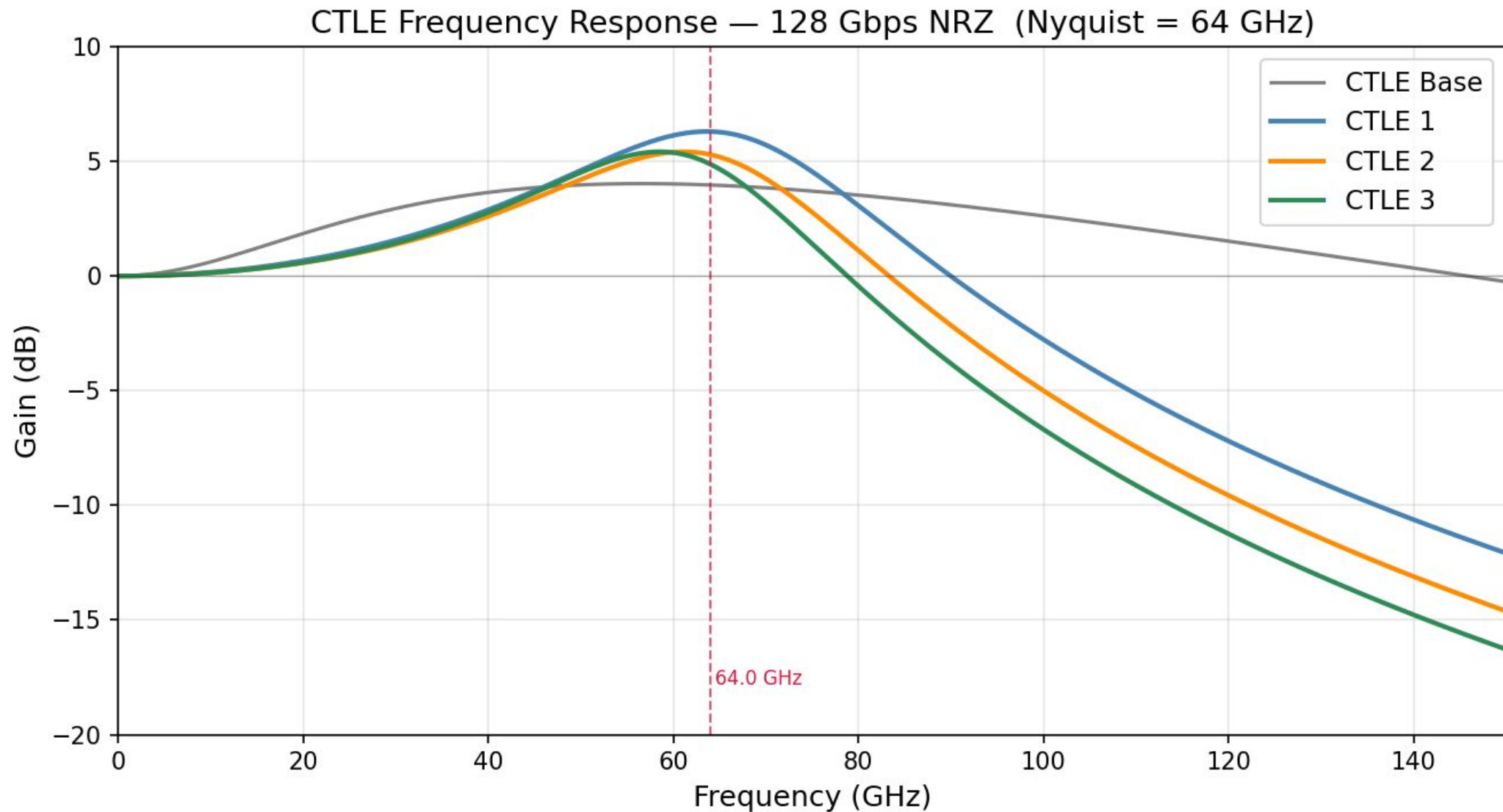




Backup



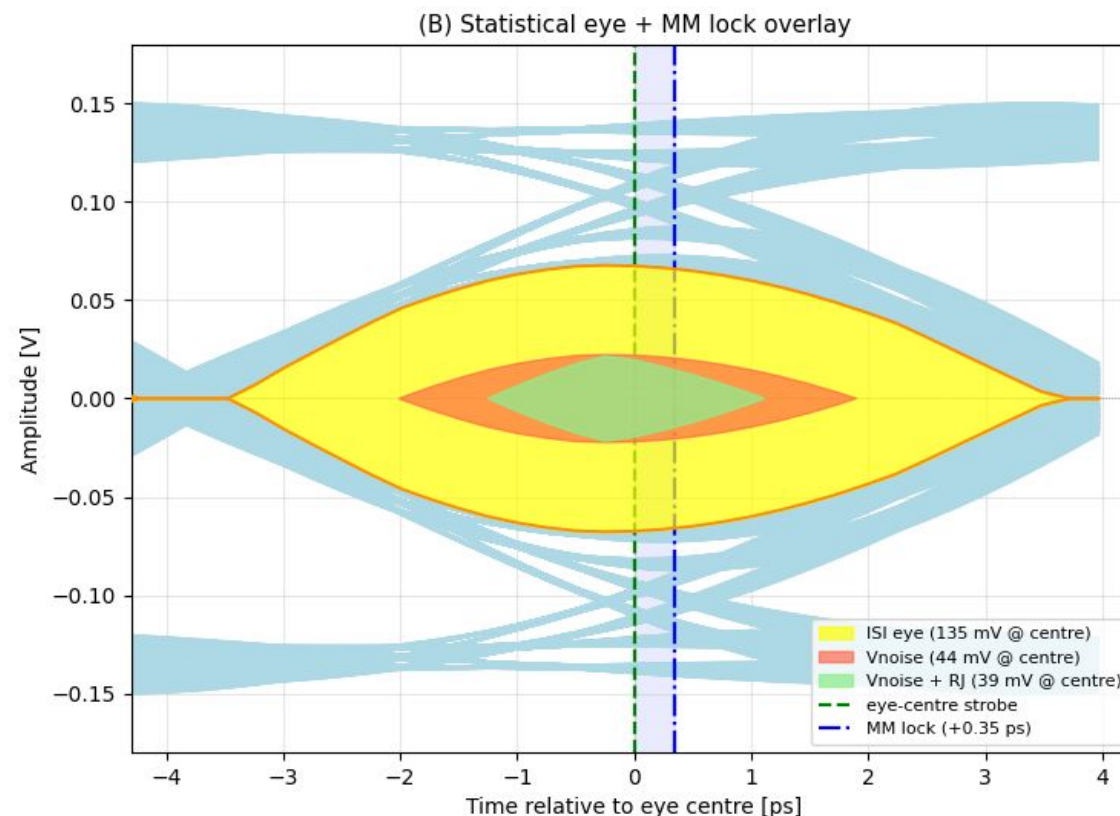
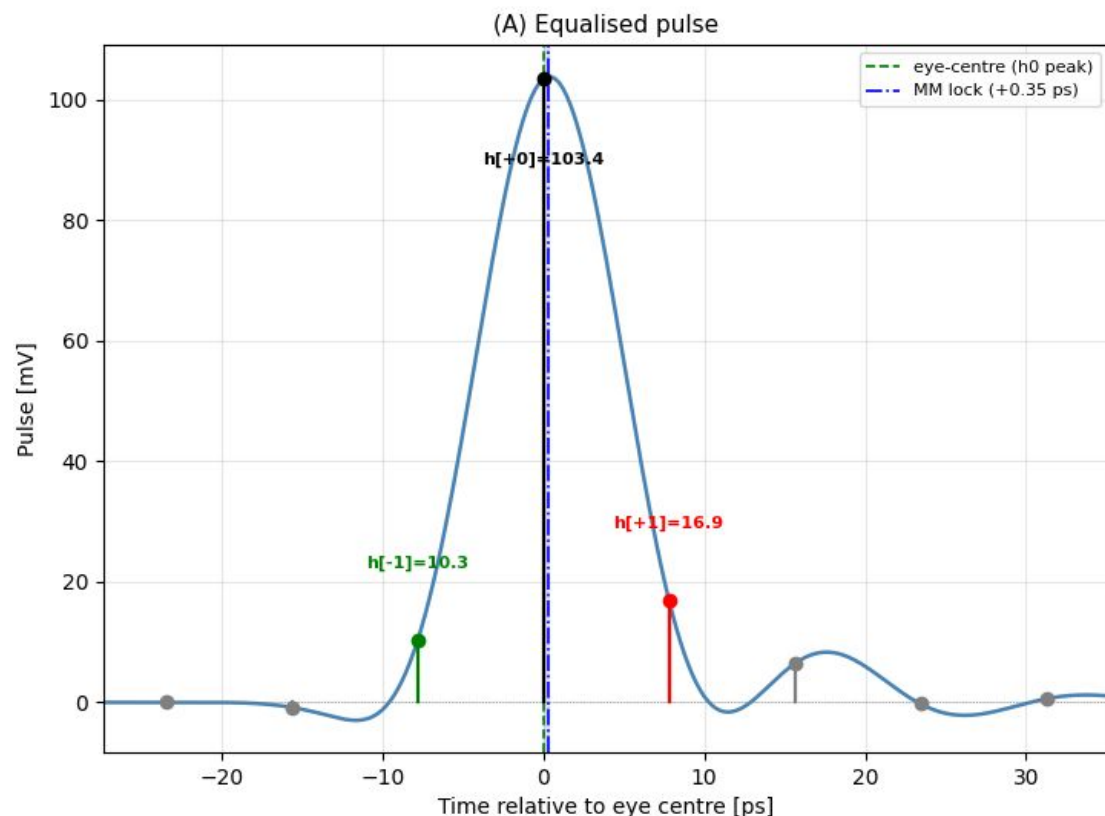
Evaluation of CTLE Responses for 128 Gb/s NRZ Operation



CTLE2 Example: Eye Opening at 128 Gb/s NRZ

TX FIR (Pre/Main/Post): [-0.08, 0.72, -0.20]

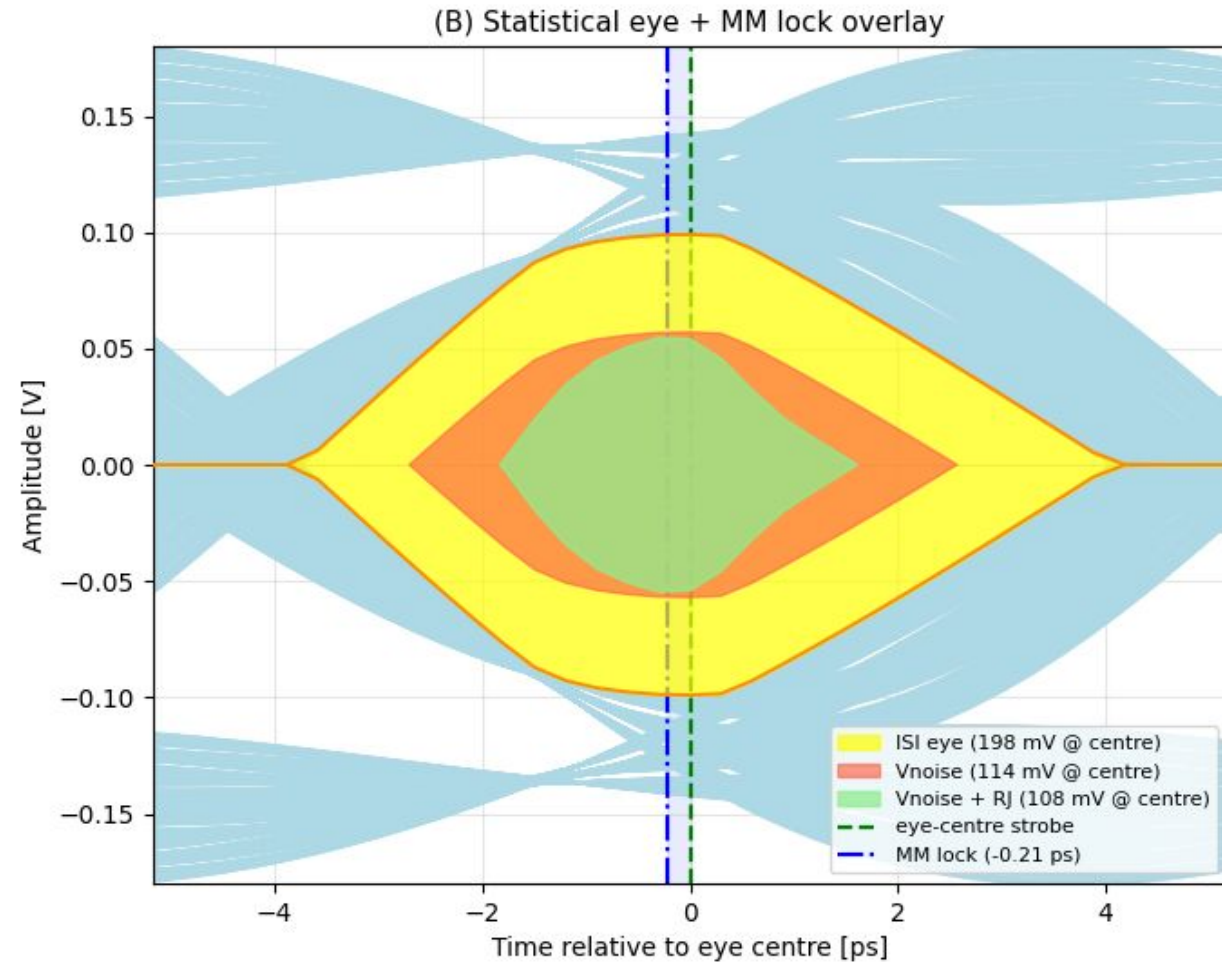
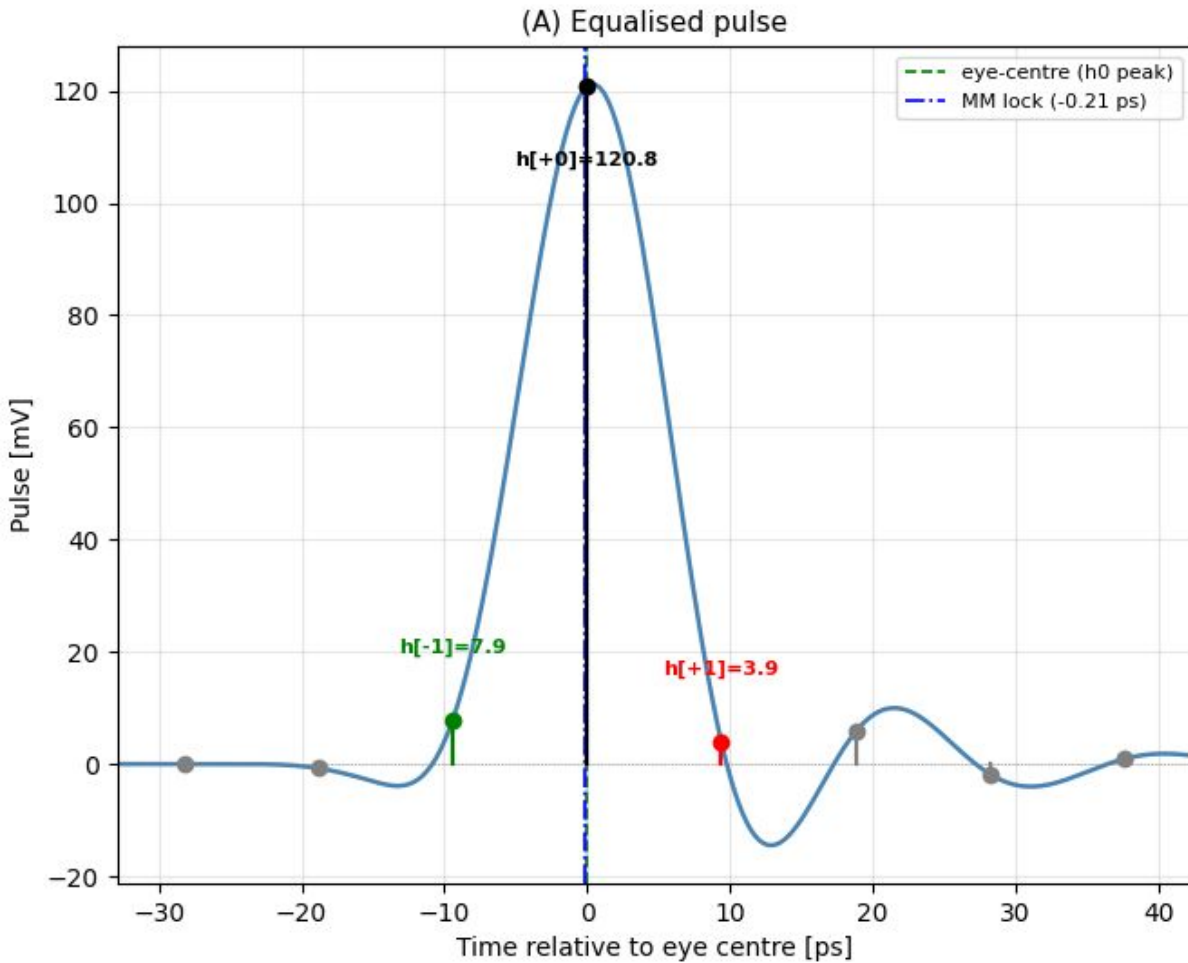
Mueller-Muller CDR | lock offset +0.35 ps (+4.5% UI) | BER=1e-12 eye: centre 39 mV → MM 32 mV



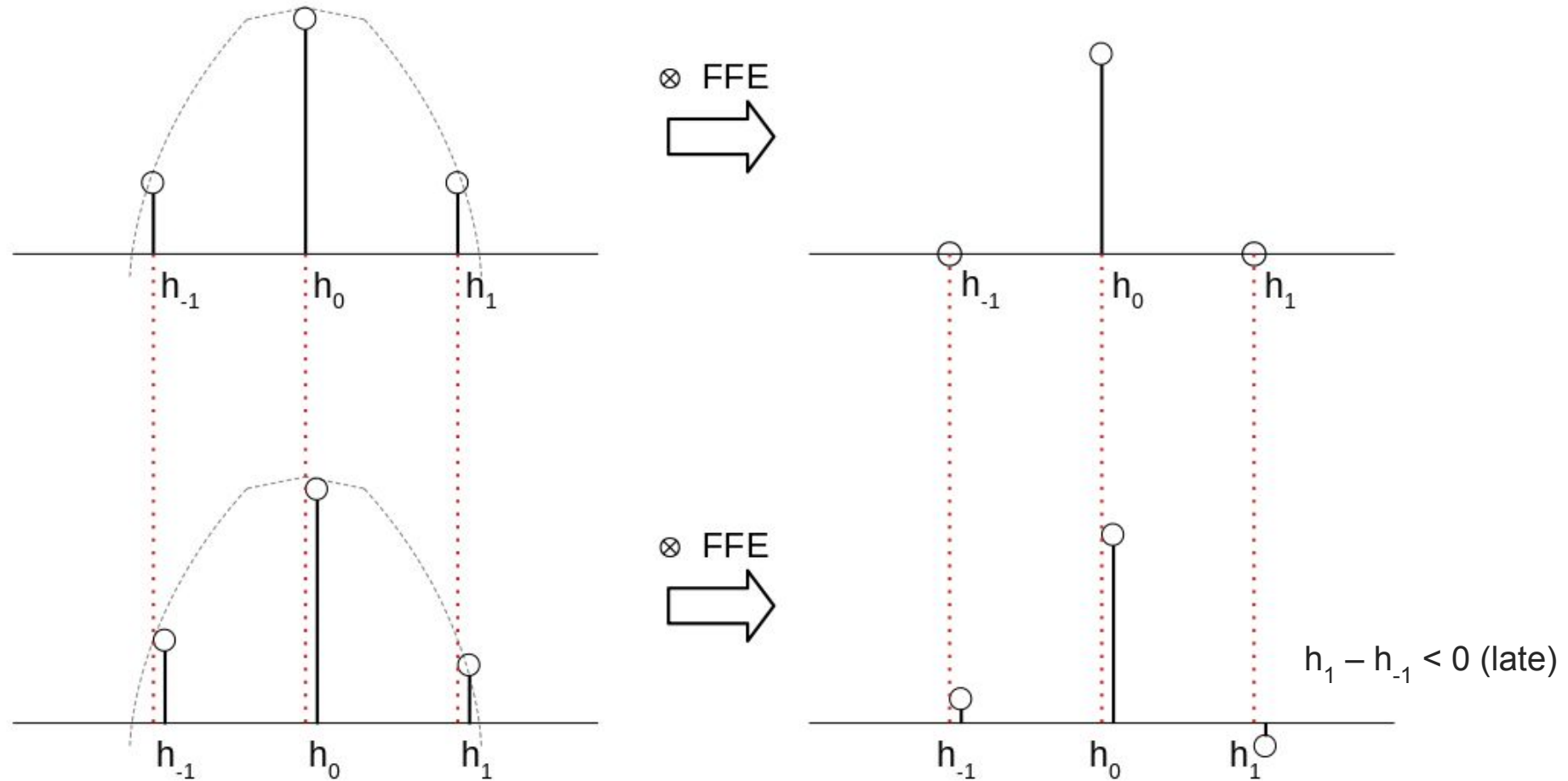
- With the applied TX FIR and the Lightmatter channel response, the evaluated CTLE configurations achieve sufficient eye opening at 128 Gb/s under the specified noise constraints.

CTLE2 Example: Eye Opening at 106.25 Gb/s NRZ

Mueller-Muller CDR | lock offset -0.21 ps (-2.3% UI) | BER=1e-12 eye: centre 108 mV → MM 109 mV



MMPD CDR with FFE



- As long as the CDR loop dynamics are faster than the FFE, the CDR loop should work.